FAMILY SEIBERG-WITTEN EQUATION ON KÄHLER SURFACE AND π_i(Symp) ON MULTIPLE-POINT BLOW UPS OF CALABI-YAU SURFACES-1009

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ABSTRACT. Let ω be a Kähler form on X, which is a torus T^4 or a $K3$ surface such that $\kappa = [\omega]$ satisfies certain irrationality condition. Let $(\tilde{X}, \tilde{\omega})$ be a Kähler form on the $n-$point blowup of (X, ω). We extend the techniques developed by Kronheimer[13] and Smirnov[27][28] and show that even dimensional higher homotopy groups of Symp$(\tilde{X}, \tilde{\omega})$ are infinitely generated.

1. INTRODUCTION

Let (X, ω) be a connected closed symplectic 4-manifold, $\mathrm{Diff}_0(X)$ be the group of diffeomorphisms isotopic to the identity, and $\mathrm{Symp}(X, \omega)$ be its symplectomorphism group. The topology of symplectomorphism groups has been studied extensively for rational or ruled surfaces [9][1][14][2][20][16]. For closed symplectic manifolds with $b^+ > 1$, there has also been substantial progress on symplectic mapping class groups [24][27][28][21][12][7]. In this paper, we focus on the higher homotopy groups of symplectomorphism groups of certain symplectic 4-manifolds with $b^+ > 1$.
Let $\mathcal{S}_\omega$ be the space of symplectic forms isotopic to ω, and denote

$$\mathrm{Symp}_{st}(X, \omega) := \mathrm{Symp}(X, \omega) \bigcap \mathrm{Diff}_0(X).$$

The Kronheimer's fibration

$$\mathrm{Symp}_{st}(X, \omega) \xrightarrow{\iota} \mathrm{Diff}_0(X) \xrightarrow{\mathrm{p}} \mathcal{S}_\omega \tag{1.1}$$

and the corresponding long exact sequence relate $K(X, \omega) := \pi_0(\mathrm{Symp}_{st}(X, \omega))$ to $\pi_1(\mathcal{S}_\omega)$. In [13], Kronheimer uses the family Seiberg-Witten theory to show that the homotopy classes of $\mathcal{S}_\omega$ can be non-trivial. Smirnov[27][28] refines this construction and proves that $K(X, \omega)$ is infinitely generated for certain Kähler forms ω on $K3$ surfaces and one-point blowups of complex two-tori. Further study of $K(X, \omega)$ is discussed in [21][7], and related applications to the study of $\pi_1(\mathrm{Diff}_0(X), id)$ appear in [26][19]. In this paper, we extend these methods to higher homotopy groups of symplectomorphism groups of multiple-point blowups of Calabi-Yau surfaces.

Let (X, ω) be a Kähler surface, where X is a $K3$ surface or a complex two-torus, and suppose that $\kappa = [\omega]$ is non-resonant, i.e. $\langle \kappa, \delta \rangle \neq 0$ for every non-zero $\delta \in H^2(X; \mathbb{Z})$, where $\langle\ ,\ \rangle$ is the intersection pairing.

Theorem A. *Let $(\tilde{X},\tilde{\omega})$ be an n-point Kähler blowup of (X,ω), let $e_1,\cdots,e_n$ be the class of exceptional divisors. Suppose the area of exceptional divisors are equal to $\lambda>0$, satisfying $2n\lambda<\sqrt{\kappa^2}$. Then*

$$\mathrm{Ker}(\iota_{j,*}:\pi_j(\mathrm{Symp}_{st}(\tilde{X},\tilde{\omega}))\to\pi_j(\mathrm{Diff}_0(\tilde{X})))$$

is infinitely generated for

$$j=2,4,\cdots,2n-2\quad \textit{when } X=T^4,\qquad j=2,4,\cdots,2n\quad \textit{when } X \textit{ is a } K3 \textit{ surface},$$

where $\iota_{j,*}$ is induced by inclusion ι.

Remark 1. Following the work in [28], when $X=T^4$ and $\tilde{\omega}$ is the Kähler form in Theorem A, $K(\tilde{X},\tilde{\omega})$ is also infinitely generated.

Now we sketch the idea of proof. Set $K=\sum_{i=1}^n e_i$, which is the canonical class of $\tilde{\omega}$. Let $\Delta_{X,k}$ be the infinite set of integral classes δ satisfying

$$(k-1)\lambda<\langle\delta,\kappa\rangle<k\lambda,$$

where δ is primitive with $\delta^2=0$ for tori, and $\delta^2=-2$ for $K3$ surfaces. Set $\epsilon_\delta=\delta-\sum_{i=1}^k e_i$, we extend Smirnov's q-invariants to homomorphisms

$$q_{\epsilon_\delta}=Q_{\epsilon_\delta}-Q_{K-\epsilon_\delta}:\pi_{2r-1}(\mathcal{S}_{\tilde{\omega}})\longrightarrow\mathbb{Z}_2,$$

where $r=k$ for T^4, $r=k+1$ for K3 surface, Q-invariants are the Kronheimer's invariants. Let

$$\cdots\xrightarrow{\iota_*}\pi_{2r-1}(\mathrm{Diff}_0(\tilde{X}))\xrightarrow{\mathrm{p}_*}\pi_{2r-1}(\mathcal{S}_{\tilde{\omega}})\xrightarrow{\partial_*}\pi_{2r-2}(\mathrm{Symp}_{st}(\tilde{X},\tilde{\omega}))\to\cdots$$

be the long exact sequence corresponding to Kronheimer's fibration (1.1). Each q_{ϵ_δ} vanishes on the image of p_*. For every finite subset $F\subset\Delta_{X,k}$, we shall prove that

$$(q_{\epsilon_\delta})_{\delta\in F}:\pi_{2r-1}(\mathcal{S}_{\tilde{\omega}})\longrightarrow\bigoplus_{\delta\in F}\mathbb{Z}_2$$

is surjective. These maps descend to $\mathrm{Ker}(\iota_{2r-2,*})$, which is therefore not finitely generated.

We briefly explain the construction of the spherical families used to prove these surjectivity statements.

Following [27][28], we begin with a disk D_δ of complex structures of X on which δ is represented by a holomorphic curve only in the central fiber. When X is a K3 surface, we consider the relative configuration space of n ordered blowup centres over D_δ, and the locus $\mathcal{H}_\delta$ where the first k centers lie on such a curve. When $X=T^4$, we consider corresponding spaces where the center of the exceptional divisor in e_i is fixed by a specific section. The corresponding locus has complex codimension

$$r=k\quad\text{for tori},\qquad r=k+1\quad\text{for } K3 \text{ surfaces}.$$

Let $B_\delta\cong D^{2r}$ be a small disk family intersecting $\mathcal{H}_\delta$ transversally once, it parametrizes a family $\{\tilde{\omega}_q|q\in B_\delta\}$ of Kähler forms such that

$$[\tilde{\omega}_q]=\kappa-\lambda\sum_{i=1}^n e_i,\quad\forall q\in S_\delta=\partial B_\delta\cong S^{2r-1}.$$

Then $\{\tilde{\omega}_q | q \in S_\delta\}$ defines an element $\alpha_\delta \in \pi_{2r-1}(\mathcal{S}_{\tilde{\omega}})$. We show that

$$q_{\epsilon_{\delta'}}(\alpha_\delta) = \begin{cases} 1, & \delta' = \delta, \\ 0, & \delta' \neq \delta. \end{cases}$$

In **Section 2**, we introduce preliminaries about Kähler geometry and period domains. In **Section 3**, we generalize the family Seiberg-Witten invariants and prove a regularity theorem. In **Section 4**, we construct the spherical families of symplectic forms and prove the main theorem.

Acknowledgement. The author would like to greatly thank Jianfeng Lin for the suggestion to study the topology of symplectomorphism group and for a lot of helpful discussion during the research. I'm also very grateful to my advisor, Tianjun Li, for a lot of helpful advice about the paper and help in symplectic geometry and gauge theory. I'd like to thank Gleb Smirnov for kindly explaining his paper to me. I would like to thank Alexander Voronov and Weiwei Wu for helpful discussion about complex geometry. And I would like to thank Hokuto Konno and Jun Li for their interest in this paper.

2. Kähler classes, period domains, and parameter spaces of blowing up on X

Throughout this section, X is a $K3$ surface or a complex torus T^4. We identify curve classes with their Poincaré duals when taking intersection products.

2.1. **The Kähler cone and point blowups.** Write $\mathcal{K}(Y) \subset H^{1,1}(Y;\mathbb{R})$ for the Kähler cone of a compact Kähler surface Y, and V_Y^+ for the component of $\{a \in H^{1,1}(Y;\mathbb{R}) | a^2 > 0\}$ containing a Kähler class. By [6, Theorem 0.1], we have

$$\mathcal{K}(Y) = \{a \in V_Y^+ | a \cdot D > 0 \text{ for every reduced irreducible curve } D \subset Y\}. \tag{2.1}$$

Let $\pi : \widetilde{X} \to X$ be the blowup at distinct points $x_1, \ldots, x_n$. Write E_i for the corresponding exceptional curves and set $e_i = \mathrm{PD}[E_i]$, so that $e_i^2 = -1$. For $\kappa \in \mathcal{K}(X)$ and $\boldsymbol{\lambda} = (\lambda_1, ..., \lambda_n)$ with $\lambda_i > 0$, put $\alpha_{\boldsymbol{\lambda}} = \pi^*\kappa - \sum_{i=1}^n \lambda_i e_i$. If $C \subset X$ is reduced and irreducible, set $m_i = \mathrm{mult}_{x_i} C$, with $m_i = 0$ when $x_i \notin C$. Its proper transform satisfies

$$\mathrm{PD}[\widetilde{C}] = \pi^*\mathrm{PD}[C] - \sum_i m_i e_i, \qquad \alpha_{\boldsymbol{\lambda}} \cdot \widetilde{C} = \kappa \cdot C - \sum_i \lambda_i m_i. \tag{2.2}$$

Lemma 2.1. *The class $\alpha_{\boldsymbol{\lambda}}$ is Kähler if and only if*

$$\kappa^2 > \sum_i \lambda_i^2, \qquad \kappa \cdot C > \sum_i \lambda_i \,\mathrm{mult}_{x_i} C \quad \textit{for every reduced irreducible } C \subset X. \tag{2.3}$$

Proof. Since $\alpha_{\boldsymbol{\lambda}}^2 = \kappa^2 - \sum_i \lambda_i^2$, If $\alpha_{\boldsymbol{\lambda}}$ is a Kähler class, then $\alpha_{\boldsymbol{\lambda}}^2 = \kappa^2 - \sum_i \lambda_i^2 > 0$ and $\alpha_{\boldsymbol{\lambda}} \cdot \widetilde{C} = \kappa \cdot C - \sum_i \lambda_i \,\mathrm{mult}_{x_i} C > 0$ for the proper transform of every reduced irreducible curve $C \subset X$. Then (2.3) follows.

Now suppose that $\alpha_{\boldsymbol{\lambda}}$ satisfies (2.3). Let $\boldsymbol{\lambda}' = (\lambda_1', ..., \lambda_n')$ such that λ_i' is sufficiently small for each i, we can see that $\alpha_{\boldsymbol{\lambda}'}$ is a Kähler class. For any $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n)$, we have

$$\alpha_{\boldsymbol{\lambda}} \cdot \alpha_{\boldsymbol{\lambda}'} = \kappa^2 - \sum_i \lambda_i \lambda_i' > \kappa^2 - \sum_i \lambda_i^2 > 0,$$

thus $\alpha_{\boldsymbol{\lambda}}$ is always contained in $V^+_{\widetilde{X}}$.

If $D \subset \widetilde{X}$ is an irreducible curve, then $D = E_i$ or D is the proper transform of some curve C on X. $\alpha_{\boldsymbol{\lambda}} \cdot E_i = \lambda_i > 0$. Then the Lemma follows from [Theorem 0.1, [6]]. □

Lemma 2.2. *Let $\pi : \widetilde{X} \to X$ be the blowup at distinct points $x_1, \dots, x_n$, with exceptional divisors $E_1, \dots, E_n$. Write $e_i = \mathrm{PD}[E_i]$ and set*

$$\alpha_{\boldsymbol{\lambda}} = \pi^*\kappa - \sum_{i=1}^n \lambda_i e_i, \qquad S = \sum_{i=1}^n \lambda_i, \qquad \ell = \Big(\sum_{i=1}^n \lambda_i^2\Big)^{1/2}, \tag{2.4}$$

where $\lambda_i > 0$ and $2S < \sqrt{\kappa^2}$. Then

$$\alpha_{\boldsymbol{\lambda}} \in V^+_{\widetilde{X}}, \qquad \alpha_{\boldsymbol{\lambda}} \cdot E_i = \lambda_i > 0. \tag{2.5}$$

If $\alpha_{\boldsymbol{\lambda}} \notin \mathcal{K}(\widetilde{X})$, or equivalently, there is a proper transform $\widetilde{C}$ of a reduced irreducible curve $C \subset X$ such that $\alpha_{\boldsymbol{\lambda}} \cdot [\widetilde{C}] \leqslant 0$, then

$$-c_X \leqslant C^2 \leqslant 0, \qquad 0 < \kappa \cdot [C] \leqslant S + \ell\sqrt{c_X}, \qquad c_X = \begin{cases} 2, & X \text{ is a K3 surface,} \\ 0, & X \text{ is a complex two-torus.} \end{cases} \tag{2.6}$$

Proof. Since $K_X = 0$, the adjunction formula and the genus formula for a point blowup imply that

$$p_a(C) = 1 + \frac{C^2}{2}, \qquad p_a(\widetilde{C}) = 1 + \frac{C^2}{2} - \frac{1}{2}\sum_{i=1}^n m_i(m_i - 1),$$

where $m_i = \mathrm{mult}_{x_i} C$. Since $p_a(\widetilde{C}) \geqslant 0$ and it's bigger than zero when X is T^4, we have

$$C^2 \geqslant -c_X, \qquad \sum_{i=1}^n m_i(m_i - 1) \leqslant C^2 + c_X.$$

Suppose that $C^2 > 0$. Since C^2 is even, we have $C^2 \geqslant 2$. For $m_i \geqslant 2$,

$$m_i^2 \leqslant 2m_i(m_i - 1) \leqslant 2(C^2 + c_X) \leqslant 4C^2.$$

Thus $m_i \leqslant 2\sqrt{C^2}$ for every i, which also holds when $m_i = 0$ or 1. The Hodge index theorem implies that

$$\alpha_{\boldsymbol{\lambda}} \cdot [\widetilde{C}] = \kappa \cdot [C] - \sum_{i=1}^n \lambda_i m_i \geqslant \big(\sqrt{\kappa^2} - 2S\big)\sqrt{C^2} > 0,$$

contrary to the hypothesis. Hence $C^2 \leqslant 0$.

Using $m_i \leqslant 1 + \sqrt{m_i(m_i - 1)}$ and Cauchy–Schwarz, we get

$$\kappa \cdot [C] \leqslant \sum_{i=1}^n \lambda_i m_i \leqslant S + \ell\Big(\sum_{i=1}^n m_i(m_i - 1)\Big)^{1/2} \leqslant S + \ell\sqrt{C^2 + c_X} \leqslant S + \ell\sqrt{c_X}.$$

This proves (2.6). Finally, (2.5) follows from the proof of Lemma 2.1. □

2.2. **Period domains of K3 surfaces and complex two-tori.** Following the method developed by Smirnov[27][28], we use period domains to get parameter spaces of the complex structures of a $K3$ surface or a complex two-torus.

The following lattices are isomorphic to the second cohomology groups of X:

$$\Lambda = \begin{cases} 3H \oplus 2(-E_8), & \text{for K3 surfaces,} \\ \bigwedge^2 \Gamma \cong 3H, & \text{for complex two-tori,} \end{cases}$$

where $H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and Γ is an oriented free abelian group of rank four. The orientation identifies $\bigwedge^4 \Gamma$ with $\mathbb{Z}$, so exterior product defines the intersection pairing on $\bigwedge^2 \Gamma$. Write $\Lambda_{\mathbb{F}} = \Lambda \otimes_{\mathbb{Z}} \mathbb{F}$ for $\mathbb{F} = \mathbb{R}, \mathbb{C}$. For a torus, we choose an orientation-preserving marking $\eta_X : H^1(X;\mathbb{Z}) \to \Gamma$ and retain the induced marking

$$\mu_X = \bigwedge^2 \eta_X : H^2(X;\mathbb{Z}) = \bigwedge^2 H^1(X;\mathbb{Z}) \longrightarrow \Lambda.$$

For a K3 surface, a marking is an isometry $\mu_X : H^2(X;\mathbb{Z}) \to \Lambda$.

For either surface, $H^{2,0}(X,\mathbb{C})$ is a complex line. Its image

$$\mathrm{Per}(X,\mu_X) := \mu_X\big(H^{2,0}(X)\big) \in \mathbb{P}(\Lambda_{\mathbb{C}})$$

is the *period point* of the marked surface. *The period domain* of X is defined as follows.

$$\mathcal{M}_X = \big\{[\sigma] \in \mathbb{P}(\Lambda_{\mathbb{C}}) : \sigma\cdot\sigma = 0, \quad \sigma\cdot\overline{\sigma} > 0\big\}. \tag{2.7}$$

We write $u = [\sigma]$ for a period point and $\ell_u = \mathbb{C}\sigma$, and abbreviate $\mathcal{M}_X$ to $\mathcal{M}$ when the surface type is understood. In a holomorphic family $\mathcal{X} \to B$ equipped with locally constant markings, these period points define a holomorphic map from B to $\mathcal{M}$, called the *period map*.

Let (X,μ_X) be a marked $K3$ surface or T^4 with period $u_0 \in \mathcal{M}_X$. For a K3 surface, the semi-universal deformation of X has a smooth base of complex dimension 20, and period map of the deformation family is a local biholomorphism by [Chapter 6, Corollary 2.7 and Proposition 2.8, [10]]. Thus its base can be identified with a sufficiently small neighbourhood U of u_0 in $\mathcal{M}$. For tori, the marked family and the period isomorphism in [Section 7, Theorems 7–8, [28]], following [25], give the same conclusion, with a four-dimensional base together with a holomorphic zero section. In either case, there is a proper holomorphic submersion

$$p : \mathcal{X}_U \longrightarrow U, \qquad X_u := p^{-1}(u), \qquad \mathrm{Per}(X_u,\mu_u) = u \tag{2.8}$$

for a sufficiently small neighborhood U of u_0.

Fix $\kappa \in \Lambda_{\mathbb{R}}$ with $\kappa^2 > 0$. To keep this class of type $(1,1)$, we restrict the period to

$$\overline{\mathcal{M}}_\kappa = \{u \in \mathcal{M} : \kappa \perp \ell_u\}. \tag{2.9}$$

The pairing on $\kappa^\perp$ has signature $(2,19)$ for K3 surfaces and $(2,3)$ for tori. Consequently, $\overline{\mathcal{M}}_\kappa$ has two contractible connected components; see [27, Section 8] and [28, Section 7].

Let (X, μ_X) be a marked surface such that $\mu_X^{-1}(\kappa) \in \mathcal{K}(X)$, and denote the component containing its period by $\overline{\mathcal{M}}_\kappa^+$. For $a \in \Lambda$, write

$$H_a := \{u \in \mathcal{M} : a \perp \ell_u\}. \tag{2.10}$$

Lemma 2.3. *Set $S(X,\kappa)$ to be $\{a \in \Lambda | a^2 = -2, \langle \kappa, a\rangle = 0\}$, then the κ-polarized period domain*

$$\mathcal{M}_\kappa = \begin{cases} \overline{\mathcal{M}}_\kappa^+ \setminus \bigcup_{a \in S(X,\kappa)} H_a, & \text{for K3 surfaces,} \\ \overline{\mathcal{M}}_\kappa^+, & \text{for complex two-tori.} \end{cases} \tag{2.11}$$

is a fine moduli space of marked surfaces for which κ is Kähler, in the sense that

$$p_\kappa : \mathcal{X}_\kappa \longrightarrow \mathcal{M}_\kappa$$

is a holomorphic submersion, and there is a smooth family $\{\omega_u | u \in \mathcal{M}_\kappa\}$ of Kähler forms on X_u such that $[\omega_u] = \kappa$. When $X = T^4$, there is a zero section $Z : \mathcal{M}_\kappa \to \mathcal{X}_\kappa$.

Proof. See [27, Section 7] [28, Section 7]. □

By the definition of non-resonance, we have the following result.

Lemma 2.4. *When X is a K3 surface and κ is non-resonant, $\mathcal{M}_\kappa = \overline{\mathcal{M}}_\kappa^+$.*

Lemma 2.5. *When κ is non-resonant, $\mathcal{M}_\kappa = \overline{\mathcal{M}}_\kappa^+$ is contractible.*

Proof. The contractibility of $\overline{\mathcal{M}}_\kappa^+$ follows from [4][27]. See also [Lemma 3.6, [7]]. □

Now we introduce some sub-domains of $\mathcal{M}_\kappa$, related to parameter space of multiple-point blowup of X, with prescribed blowup sizes.

Set

$$\mathrm{Conf}_n(X_u) = \{(x_1, \ldots, x_n) \in X_u^n | x_i \neq x_j \text{ if } i \neq j\}.$$

For $\mathbf{x} \in \mathrm{Conf}_n(X_u)$, let $\pi_{u,\mathbf{x}} : \widetilde{X}_{u,\mathbf{x}} \to X_u$ be the natural projection, and write $e_i = \mathrm{PD}[E_i]$ for the exceptional classes. With the identification $\mu_u : H^2(X_u; \mathbb{Z}) \cong \Lambda$, κ can be considered as a class on X_u.

Definition 2.6. Let $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n)$ with $\lambda_i > 0$ and $\sum_i \lambda_i^2 < \kappa^2$. For distinct points $\mathbf{x} = (x_1, \ldots, x_n) \in \mathrm{Conf}_n(X_u)$, let $\tilde{X}_{u,x_1,\ldots,x_n} \to X_u$ be the blowdown, let e_i be the corresponding exceptional classes and set $\alpha_{u,\mathbf{x}} = \pi_{u,\mathbf{x}}^* \kappa - \sum_{i=1}^n \lambda_i e_i$. Define

$$\mathcal{M}_{\kappa,\boldsymbol{\lambda}} = \Big\{u \in \mathcal{M}_\kappa : \pi_{u,\mathbf{x}}^* \kappa - \sum_i \lambda_i e_i \in \mathcal{K}(\mathrm{Bl}_{x_1,\ldots,x_n} X_u),\ \forall \mathbf{x} \in \mathrm{Conf}_n(X_u)\Big\}. \tag{2.12}$$

For a single blowup of size S, write $\mathcal{M}_{\kappa,S}$.

By Lemma 2.1, then $\alpha_{u,\mathbf{x}} \in \mathcal{K}(\widetilde{X}_{u,\mathbf{x}})$ is equivalent to

$$\kappa \cdot [C] > \sum_{i=1}^n \lambda_i \,\mathrm{mult}_{x_i} C \text{ for every reduced irreducible curve } C \subset X_u.$$

Lemma 2.7. $\mathcal{M}_{\kappa,\sum \lambda_i} \subseteq \mathcal{M}_{\kappa,\lambda_1,\ldots,\lambda_n}$.

Proof. $u \in \mathcal{M}_{\kappa,\sum_i \lambda_i}$ implies that $\kappa \cdot [C] > (\sum_{i=1}^n \lambda_i) \operatorname{mult}_x C$ for every reduced irreducible curve $C \subset X_u$. Let $x = x_i$ for each i, and suppose that $\operatorname{mult}_{x_1} C \geqslant \operatorname{mult}_{x_i} C$ for each i. Then we have

$$\kappa \cdot [C] > (\sum_{i=1}^n \lambda_i) \operatorname{mult}_{x_1} C \geqslant \sum_{i=1}^n \lambda_i \operatorname{mult}_{x_i} C$$

for each irreducible C in X_u. □

Lemma 2.8. *Suppose that* $S := \sum_{i=1}^n \lambda_i < \frac{1}{2}\sqrt{\kappa^2}$, *then* $\mathcal{M}_{\kappa,\lambda_1,\ldots,\lambda_n}$ *is nonempty, dense, and path connected.*

Proof. To exclude the possible obstructions from curves of nonpositive square, set

$$\mathcal{I}_X(S) = \begin{cases} \{a \in \Lambda | a^2 \in \{-2, 0\},\ 0 < \kappa \cdot a \leqslant 2S\}, & X \text{ a K3 surface}, \\ \{a \in \Lambda | a^2 = 0,\ 0 < \kappa \cdot a \leqslant S\}, & X \text{ a torus}, \end{cases}$$

and define

$$\mathcal{G}_{\kappa,S} = \mathcal{M}_\kappa \backslash \bigcup_{a \in \mathcal{I}_X(S)} H_a.$$

For each $a \in \mathcal{I}_X(S)$, $H_a^\kappa := H_a \bigcap \mathcal{M}_\kappa$ is a smooth complex hypersurface. By [Lemma 12, [28]], hyperplanes in $\bigcup_{a \in \mathcal{I}_X(S)} H_a^\kappa$ are locally finite. By [Lemma 5,[27]], $\bigcup_{\{a \in \mathcal{I}_X(S) | a^2 = -2\}} H_a^\kappa$ is locally finite. And a similar result holds for $\bigcup_{\{a \in \mathcal{I}_X(S) | a^2 = 0\}} H_a^\kappa$ by [Lemma 12, [28]], see also [7]. As a result, $\mathcal{G}_{\kappa,S}$ is open and dense in $\mathcal{M}_\kappa$. Since $\mathcal{M}_\kappa$ is path connected, the complement of countably many complex hyperplanes in $\mathcal{M}_\kappa$ is path connected.

Let $u \in \mathcal{G}_{\kappa,S}$ and let $C \subset X_u$ be an irreducible curve s.t. $C^2 \leqslant 0$.

By the adjunction formula on $K3$ surface, $C^2 \in \{-2, 0\}$. Let $\widetilde{C}$ be the proper transform of C under the blowup at x. Let m be the multiplicity if $x \in C$, and set $m = 0$ if $x \notin C$. Then

$$p_a(\widetilde{C}) = 1 + \frac{C^2}{2} - \frac{m(m-1)}{2} \geqslant 0,$$

thus $m(m-1) \leqslant C^2 + 2 \leqslant 2$, $m \leqslant 2$. By the definition of $\mathcal{I}_X(S)$, we have

$$\kappa \cdot C > 2S \geqslant S \cdot m.$$

For a complex two-torus, an irreducible curve C with $C^2 \leqslant 0$ is a smooth elliptic curve of square zero. Hence $m \leqslant 1$, while the definition of $\mathcal{I}_X(S)$ implies

$$\kappa \cdot C > S \geqslant Sm.$$

Now let $C \subset X_u$ be irreducible curve with $C^2 > 0$. By Lemma 2.2, $\kappa \cdot [C] > S \operatorname{mult}_x C$ for every reduced irreducible curve $C \subset X_u$ and every $x \in X_u$. In conclusion, if $u \in \mathcal{G}_{\kappa,S}$, then $\pi_{u,x}^* \kappa - Se$ is a Kähler class on $\tilde{X}_{u,x}$ for each $x \in X_u$, where e is the exceptional class of $\tilde{X}_{u,x}$ for each $x \in X_u$, thus $\mathcal{G}_{\kappa,S} \subseteq \mathcal{M}_{\kappa,S}$. By Lemma 2.7, we have

$$\mathcal{G}_{\kappa,S} \subseteq \mathcal{M}_{\kappa,S} \subseteq \mathcal{M}_{\kappa,\lambda_1,\ldots,\lambda_n}.$$

Finally, let $u \in \mathcal{M}_{\kappa,\lambda_1,\ldots,\lambda_n}$. By local finiteness of hyperplanes parametrized by $\mathcal{I}_X(S)$, we can choose a path $\gamma : [0, 1] \to \mathcal{M}_\kappa$ such that

$$\gamma(0) = u, \qquad \gamma(t) \in \mathcal{G}_{\kappa,S} \quad (0 < t \leqslant 1).$$

Since u and $\mathcal{G}_{\kappa,S}$ both belong to $\mathcal{M}_{\kappa,\lambda_1,\ldots,\lambda_n}$, the whole path lies in this locus. As $\mathcal{G}_{\kappa,S}$ is dense and path connected, the claimed nonemptiness, density, and path connectedness follow.

□

2.3. **Local parameter families for multiple-point blowups.** In this section, we construct some parameter families of Kähler structures on the blowup of X, under the nonresonance assumtion of κ.

By Lemma 2.5, the holomorphic family

$$p_\kappa : \mathcal{X}_\kappa \to \mathcal{M}_\kappa$$

is smoothly trivial. From now on, we fix a specific smooth trivialization of $\mathcal{X}_\kappa$. Thus κ is a Kähler class on every fiber X_u. When $X = T^4$, there is a natural holomorphic section obtained from the global section Z. Here, $\dim_{\mathbb{C}} \mathcal{M}_\kappa = 19$ for K3 surfaces and 3 for tori.

Definition 2.9. We define the relative configuration space of $\mathcal{X}_\kappa$

$$r : \overline{\mathcal{P}}_\kappa := \mathrm{Conf}_n(\mathcal{X}_\kappa/\mathcal{M}_\kappa) = \left(\mathcal{X}_\kappa \underset{\mathcal{M}_\kappa}{\times} \cdots \underset{\mathcal{M}_\kappa}{\times} \mathcal{X}_\kappa\right) \setminus \bigcup_{i<j} \{x_i = x_j\} \to \mathcal{M}_\kappa, \quad (u, \mathbf{x}) \to u \tag{2.13}$$

then $r^{-1}(u) = \mathrm{Conf}_n(X_u)$. Recall the canonical zero section $Z : \mathcal{M}_\kappa \to \mathcal{X}_\kappa$ when X is a complex torus, then set

$$\mathcal{P}_\kappa = \begin{cases} \overline{\mathcal{P}}_\kappa, & \text{for K3 surfaces,} \\ \{(u, x_1, \ldots, x_n) \in \overline{\mathcal{P}}_\kappa : x_1 = Z(u)\}, & \text{for tori.} \end{cases} \tag{2.14}$$

Remark 2. $\mathcal{P}_\kappa$ parametrizes configurations modulo translation in the torus case.

The complex dimensions of $\overline{\mathcal{P}}_\kappa$ and $\mathcal{P}_\kappa$ are respectively $2n+3$ and $2n+1$ for tori, and both are $2n+19$ for K3 surfaces.

With the smooth trivialization $\mathcal{X}_\kappa \simeq \mathcal{M}_\kappa \times X_{u_0}$, chosen to send Z to the constant zero section for tori, we have identifications

$$\begin{aligned} &\overline{\mathcal{P}}_\kappa \simeq \mathcal{M}_\kappa \times \mathrm{Conf}_n(X_{u_0}), \\ &\mathcal{P}_\kappa \simeq \mathcal{M}_\kappa \times \mathrm{Conf}_{n-1}(X_{u_0} \backslash \{0\}) \quad \text{for tori,} \end{aligned} \tag{2.15}$$

Set $\mathcal{Y} = \overline{\mathcal{P}}_\kappa \times_{\mathcal{M}_\kappa} \mathcal{X}_\kappa \longrightarrow \overline{\mathcal{P}}_\kappa$, each marked point $\mathbf{x} = (x_1, ..., x_n) \in \mathrm{Conf}_n(X_u)$ defines n tautological holomorphic sections $s_i(q) = (q, x_i)$. Since images of s_i are disjoint, we may blow up all of them:

$$\Pi : \widetilde{\mathcal{X}} = \mathrm{Bl}_{\coprod_i s_i(\overline{\mathcal{P}}_\kappa)} \mathcal{Y} \longrightarrow \mathcal{Y}, \qquad \widetilde{p} : \widetilde{\mathcal{X}} \longrightarrow \overline{\mathcal{P}}_\kappa. \tag{2.16}$$

$\widetilde{p}$ is a proper holomorphic submersion with fiber $\widetilde{X}_q = \mathrm{Bl}_{x_1,\ldots,x_n} X_u$. Let $\pi_q : \widetilde{X}_q \to X_u$ be its blowdown and $e_i = \mathrm{PD}[E_{i,q}]$ the exceptional classes. $\widetilde{\mathcal{X}}$ is a homological trivial family, we can identify the cohomology local system with $\Lambda \oplus \bigoplus_{i=1}^n \mathbb{Z} e_i$. From now on, we also denote the pullback $\pi_q^* \kappa$ by κ. Thus $\kappa - \sum_i \lambda_i e_i$ denotes the class $\pi_q^* \kappa - \sum_i \lambda_i e_i$ on $\widetilde{X}_q$. The family over $\mathcal{P}_\kappa$ is the restriction of (2.16).

Set $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n),\ \lambda_i > 0,\ \sum_i \lambda_i^2 < \kappa^2$ and $\alpha_{\kappa,\boldsymbol{\lambda}} = \kappa - \sum_i \lambda_i e_i$. Then $\alpha_{\kappa,\boldsymbol{\lambda}}$ can be considered as a $(1,1)$-class of each $\tilde{X}_q$, and we define

$$\overline{\mathcal{P}}_{\kappa,\boldsymbol{\lambda}} = \{q \in \overline{\mathcal{P}}_\kappa : \alpha_{\kappa,\boldsymbol{\lambda}} \in \mathcal{K}(\tilde{X}_q)\}, \qquad \mathcal{P}_{\kappa,\boldsymbol{\lambda}} := \overline{\mathcal{P}}_{\kappa,\boldsymbol{\lambda}} \bigcap \mathcal{P}_\kappa \tag{2.17}$$

Set $S = \sum_{i=1}^n \lambda_i$ and suppose that $2S < \sqrt{\kappa^2}$. By the definition of $\mathcal{M}_{\kappa,\boldsymbol{\lambda}}$ and the proof of Lemma 2.8,

$$r^{-1}(\mathcal{G}_{\kappa,S}) \subseteq r^{-1}(\mathcal{M}_{\kappa,\boldsymbol{\lambda}}) \subseteq \overline{\mathcal{P}}_{\kappa,\boldsymbol{\lambda}}.$$

Intersecting with $\mathcal{P}_\kappa$ gives the corresponding inclusions in $\mathcal{P}_\kappa$.

Lemma 2.10. *$\mathcal{P}_{\kappa,\boldsymbol{\lambda}}$ is open, dense and path connected in $\mathcal{P}_\kappa$. For any two $q, q' \in \mathcal{P}_{\kappa,\boldsymbol{\lambda}}$ and Kähler forms $\tilde{\omega}_q, \tilde{\omega}_{q'}$ on $\tilde{X}_q, \tilde{X}_{q'}$ in class $\alpha_{\kappa,\boldsymbol{\lambda}}$, there is a diffeomorphism*

$$h : (\tilde{X}_q, \tilde{\omega}_q) \to (\tilde{X}_{q'}, \tilde{\omega}_{q'})$$

Proof. If $q \in \mathcal{P}_{\kappa,\boldsymbol{\lambda}}$, then there is a Kähler form $\tilde{\omega}_q$ on $\tilde{X}_q$ with $[\tilde{\omega}_q] = \alpha_{\kappa,\boldsymbol{\lambda}}$. With a local smooth trivialization of the blowup family, by the openness of tameness, there is a small neighborhood U of q in $\mathcal{P}_\kappa$ such that $\tilde{\omega}_q$ tames $\tilde{X}_{q'}$ for each $q' \in U$. For each $q' \in U$, $\kappa \in H^{1,1}(\tilde{X}_{q'}; \mathbb{R})$, thus $\alpha_{\kappa,\boldsymbol{\lambda}}$ remains of type $(1,1)$. By [Theorem 1.3, [18]], $[\tilde{\omega}_q]$ is a Kähler class of $\tilde{X}_{q'}$. As a result, $\mathcal{P}_{\kappa,\boldsymbol{\lambda}}$ is an open subset of $\mathcal{P}_\kappa$. In particular, $\mathcal{P}_{\kappa,\boldsymbol{\lambda}}$ is locally path connected.

Since $\mathcal{G}_{\kappa,S}$ is dense and path connected in $\mathcal{M}_\kappa$, $r^{-1}(\mathcal{G}_{\kappa,S}) \cap \mathcal{P}_\kappa$ is also dense and path connected in $\mathcal{P}_\kappa$. Since $r^{-1}(\mathcal{G}_{\kappa,S}) \cap \mathcal{P}_\kappa \subset \mathcal{P}_{\kappa,\boldsymbol{\lambda}}$, $\mathcal{P}_{\kappa,\boldsymbol{\lambda}}$ is dense in $\mathcal{P}_\kappa$. Together with the local path-connectedness of $\mathcal{P}_{\kappa,\boldsymbol{\lambda}}$, we get the path connectedness of $\mathcal{P}_{\kappa,\boldsymbol{\lambda}}$.

Let $q, q' \in \mathcal{P}_{\kappa,\boldsymbol{\lambda}}$ and let $\tilde{\omega}_q, \tilde{\omega}_{q'}$ be Kähler forms on $\tilde{X}_q, \tilde{X}_{q'}$, respectively. Let $I : [0,1] \to \mathcal{P}_{\kappa,\boldsymbol{\lambda}}$ be a path connecting q and q', take a smooth trivialization of the blowup family along I. Then there is a path of cohomologous Kähler forms $\{\tilde{\omega}_t | t \in [0,1]\}$ connecting $\tilde{\omega}_q$ and $\tilde{\omega}_{q'}$. Applying Moser's isotopy to the path, we get a diffeomorphism between the symplectic structures preserving the cohomology. □

3. Family Seiberg Witten invariant, Kronheimer and Smirnov's invariant

3.1. **Family Seiberg Witten equation.** Let X be a connected, closed, smooth, oriented $4-$dimensional manifold. Let b_i be $i-$th betti number of X and b^+ be the dimension of the positive subspace of $H^2(X; \mathbb{R})$.

Let $\pi : \mathcal{X} \to B$ be a smooth fiber bundle, where the fiber of the bundle is diffeomorphic to X. Define $\mathrm{T}_{\mathcal{X}/B}$ to be the vertical tangent bundle of $\mathcal{X}$, i.e. a subbundle of $T\mathcal{X}$ defined as $\ker(\pi_*)$. We choose a smooth family of metric $\{g_b\}$ on $\mathrm{T}_{\mathcal{X}/B}$, which defines a principal $\mathrm{SO}(4)$ bundle of frames, $Fr \to \mathcal{X}$. A family spin^c structure $\mathcal{L}$ is a family of lifting of the frame bundle Fr, to a principal spin$^c(4)$ bundle over $\mathcal{X}$. Given a family spin^c structure $\mathcal{L}$, we have two natural $\mathrm{U}(2)$ bundles, S^+, S^-. The natural homomorphism $\mathrm{T}^*(\mathcal{X}/B) \otimes S^+ \to S^-$ defines the Clifford homomorphism, and its adjoint defines a canonical homomorphism

$$\tau : \mathrm{End}(S^+) \to \Lambda^+ \otimes \mathbb{C}$$

Here, $\Lambda^+ := \Lambda^+ T^*_{\mathcal{X}/B}$ is the vector bundle of self-dual vertical 2-forms over $\mathcal{X}$, defined by the family of metrics $\{g_b\}$.

Let $L := \det(S^+)$. A unitary connection A_b on L_b, together with the Levi-Civita connection of g_b, determines a covariant derivative ∇_{A_b} on the spinor bundles.

Let $\mathcal{A}(L_b)$ be the space of unitary connections on L_b and set

$$\mathcal{C}_b := C^\infty(X_b, S_b^+) \times \mathcal{A}(L_b).$$

Following [17], define the configuration space to be

$$\mathcal{C} := \{(\Phi, A) = \{(\Phi_b, A_b)\}_{b\in B} \mid (\Phi_b, A_b) \in \mathcal{C}_b \text{ depends smoothly on } b\}. \tag{3.1}$$

Its irreducible part $\mathcal{C}^*$ consists of families with $\Phi_b \not\equiv 0$ on X_b for every $b \in B$.

Let $\{\eta_b \in \Omega^{2,+}_{g_b}(X_b; \mathbb{R})\}$ be a smooth family of real self-dual forms, the perturbed family Seiberg–Witten equations are

$$\begin{cases} \mathcal{D}_{A_b}\Phi_b = 0, \\ \quad F^+_{A_b} = \dfrac{1}{4}\tau(\Phi_b \otimes \Phi_b^*) + i\eta_b. \end{cases} \tag{3.2}$$

Here $\mathcal{D}_{A_b} : C^\infty(X_b, S_b^+) \to C^\infty(X_b, S_b^-)$ is the Dirac operator obtained by composing ∇_{A_b} with Clifford multiplication. A solution on X_b is reducible if $\Phi_b \equiv 0$, and irreducible otherwise.

By [13],[17], the formal dimension of Seiberg-Witten moduli space as a real manifold is:

$$d(\mathcal{L}, B) = \frac{c_1(L_b)^2 - 2\chi(X) - 3\sigma(X)}{4} + \dim_{\mathbb{R}} B, \tag{3.3}$$

where $L_b = \det(S_b^+)$.

Let $\Gamma_{\mathcal{X}}$ be the space of smooth families of metrics and perturbations

$$\Gamma_{\mathcal{X}} := \{(g, \eta) \mid g = \{g_b | b \in B\},\ \eta = \{\eta_b | b \in B\},\ \eta_b \in \Omega^{2,+}_{g_b}(X_b; \mathbb{R})\}. \tag{3.4}$$

And the moduli space of family Seiberg Witten equation solutions is defined as

$$\mathcal{M}^{\mathcal{L}}_{g,\eta} := \bigcup_{b\in B} \mathcal{M}^{\mathcal{L}_b}_{g_b,\eta_b} := \bigcup_{b\in B} \{\text{ Solutions of the family Seiberg-Witten equation at } b\}/\mathcal{G}_b \tag{3.5}$$

Here, $\mathcal{G}_b = Map(\mathcal{X}_b, U(1))$, and they form a bundle $\mathcal{G}$ of groups over B.
Let $\mathcal{H}_{g_b}$ be the space of g_b-harmonic 2-forms and identify it with $H^2(X_b; \mathbb{R})$, we use $H^+_{g_b}$ to denote the corresponding harmonic projection. By (3.2), a reducible solution occurs over b if and only if

$$H^+_{g_b}(\eta_b) + 2\pi H^+_{g_b}(c_1(L_b)) = 0.$$

Thus we define the $\mathcal{L}$-wall to be

$$\mathrm{WA}(\mathcal{L}) := \left\{(g, \eta) \in \Gamma_{\mathcal{X}} : H^+_{g_b}(\eta_b) + 2\pi H^+_{g_b}(c_1(L_b)) = 0 \text{ for some } b \in B\right\}. \tag{3.6}$$

Definition 3.1. We define $\Gamma^*_{\mathcal{L}}$ to be $\Gamma_{\mathcal{X}} \backslash \mathrm{WA}(\mathcal{L})$ and define a chamber of $\Gamma^*_{\mathcal{L}}$ to be a connected component of it.

With the chamber of the parameter family specified, we can define the family Seiberg-Witten invariant. When

$$d(\mathcal{L}, B) = \frac{c_1(L_b)^2 - 2\chi(X) - 3\sigma(X)}{4} + \dim_{\mathbb{R}} B = 0, \tag{3.7}$$

a generic parameter family (g,η) in the chosen chamber defines a compact, zero-dimensional Seiberg–Witten moduli space. We define the mod-2 family invariant by

$$\mathrm{FSW}_{\mathcal{L}}(\mathcal{X}) := \#\mathcal{M}^{\mathcal{L}}_{g,\eta} \in \mathbb{Z}_2.$$

Now we focus on the case that $B = S^n$ with $n \geqslant 2$ and $b^+(X) > 1$. Since S^n is simply connected, there are identifications $H^2(X_b;\mathbb{R}) \cong H^2(X;\mathbb{R})$. The fiberwise positive cones of X_b form a trivial bundle $S^n \times \mathbf{K}(X)$, where

$$\mathbf{K}(X) := \{a \in H^2(X;\mathbb{R}) \mid a^2 > 0\}. \tag{3.8}$$

By [17][7], for each $(g,\eta) \in \Gamma^*_{\mathcal{L}}$, there is a map

$$S^n \to \mathbf{K}(X), \qquad Pr_{\mathcal{L}}(b) := H^+_{g_b}(\eta_b) + 2\pi H^+_{g_b}(c_1(L_b))$$

The chamber of (g,η) is determined by the homotopy class of this map. Following [Lemma 2.3 and Definition 2.5, [19]], the canonical chamber is the chamber corresponding to the homotopy class of constant maps.

Set $\mathbf{K}_{a_0}(X) = \{a \in \mathbf{K}(X) | a_0 \cdot a > 0\}$. It's easy to see that $\mathbf{K}_{a_0}(X)$ is contractible for each $a_0 \in \mathbf{K}(X)$. Then we have the following Lemma.

Lemma 3.2. [21] *If $Pr_{\mathcal{L}}(b) \in \mathbf{K}_{a_0}$ for every $b \in S^n$, then (g,η) lies in the canonical chamber.*

We recall the family Seiberg–Witten invariant of a spherical family. Let $r \geqslant 1$ and let

$$f : S^{2r-1} \longrightarrow \mathrm{Diff}_0(X), \quad t \to f_t$$

be a smooth based map. Its clutching family is

$$\pi_f : \mathcal{X}_f := (D^{2r}_+ \times X) \cup_f (D^{2r}_- \times X) \longrightarrow S^{2r}, \tag{3.9}$$

where $(t,x)_+ \sim (t, f_t(x))_-$ for $t \in S^{2r-1}$.

Fix a spin^c structure $\mathfrak{s}$ on X such that

$$d(\mathfrak{s}) = \frac{c_1(\det \mathfrak{s})^2 - 2\chi(X) - 3\sigma(X)}{4} = -2r,$$

and suppose that $\mathfrak{s}$ extends to a family spin^c structure $\mathcal{L}_f$ on the vertical tangent bundle $T_{\mathcal{X}_f/S^{2r}}$.

Definition 3.3. For such a map f, define

$$\mathrm{FSW}_{\mathfrak{s}}([f]) := \mathrm{FSW}_{\mathcal{L}_f}(\mathcal{X}_f) = \#\mathcal{M}^{\mathcal{L}_f}_{g,\eta} \pmod 2 \in \mathbb{Z}_2, \tag{3.10}$$

where (g,η) is a generic parameter family in the canonical chamber.

Any two extensions of $\mathfrak{s}$ differ by twisting with a line bundle pulled back from S^{2r}, which does not change the value of $\mathrm{FSW}_{\mathfrak{s}}([f])$ by [19, Lemma 2.7]. It is an invariant of the homotopy class of f in $\mathrm{Diff}(X)$; see [19][3].

3.1.1. *Relative family Seiberg–Witten invariants on disk families.* Let $\mathfrak{s}$ be a spin^c structure of TX. Over a smoothly trivial family $X \times \mathbb{D}^n$, restriction defines a one to one correspondence between spin^c structures of $X \times \mathbb{D}^n$ and the spin^c structure of X. And we can consider the parameter (g, η) as a map from $\mathbb{D}^n$ to

$$\Gamma := \{(g,\eta) | g \in \mathrm{Met}(X), \eta \text{ is a g-self-dual } 2-\text{form}\}, \tag{3.11}$$

the space of parameters of Seiberg-Witten equations on X, and set

$$\Gamma^*_{\mathfrak{s}} := \left\{(g,\eta) \in \Gamma | H^+_g(\eta) + 2\pi H^+_g(c_1(\det \mathfrak{s})) \neq 0\right\}. \tag{3.12}$$

Set

$$Pr((g,\eta)) := H^+_g(\eta) + 2\pi H^+_g(c_1(\det \mathfrak{s})) \in \mathbf{K}(X). \tag{3.13}$$

By [17][21], Pr is a homotopy equivalence from $\Gamma^*_{\mathfrak{s}}$ to $\mathbf{K}(X)$. For $a_0 \in \mathbf{K}(X)$, set

$$\Gamma^*_{\mathfrak{s}}(a_0) := \{(g,\eta) \in \Gamma : \langle Pr(g,\eta), a_0\rangle > 0\}.$$

Lemma 3.4. *Every map $p : S^{n-1} \to \Gamma^*_{\mathfrak{s}}(a_0)$ extends to $D^n \to \Gamma^*_{\mathfrak{s}}(a_0)$. Any two such extensions are homotopic relative to the boundary through maps into $\Gamma^*_{\mathfrak{s}}(a_0)$.*

Proof. By [21, Lemma 3.1], the restriction $Pr : \Gamma^*_{\mathfrak{s}}(a_0) \to \mathbf{K}_{a_0}(X)$ is a fibration with contractible fibers. The claim follows since $\mathbf{K}_{a_0}(X)$ is contractible. □

Let $\Delta_{\mathfrak{s}} := \{(g,\eta) \in \Gamma | \mathcal{M}^{\mathfrak{s}}_{g,\eta} \neq \varnothing\}$. Let $p : S^n \longrightarrow \Gamma^*_{\mathfrak{s}}(a_0)$ be a parameter family such that $\mathrm{Im}(Pr \circ p) \subset \mathbf{K}_{a_0}(X)$ for some $a_0 \in \mathbf{K}(X)$. Suppose that $\mathrm{Im}(p) \bigcap \Delta_{\mathfrak{s}} = \varnothing$ and

$$\mathrm{d}(\mathfrak{s}) + n + 1 = 0.$$

For a generic extension $\widehat{p} : D^{n+1} \longrightarrow \Gamma^*_{\mathfrak{s}}(a_0)$ of p, as in Lemma 3.4, $\bigcup_{b \in D^{n+1}} \mathcal{M}^{\mathfrak{s}_b}_{g_b,\eta_b}$ is a compact 0-dimensional manifold, and we define

$$Q_{p,a_0,\mathfrak{s}} := \#\left(\coprod_{b \in D^{n+1}} \mathcal{M}^{\mathfrak{s}_b}_{g_b,\eta_b}\right) \pmod 2 \in \mathbb{Z}_2. \tag{3.14}$$

Different admissible extensions are homotopic relative to their boundaries, so there is a cobordism between the corresponding moduli spaces.

Lemma 3.5. *The evaluation of $Q_{p,a_0,\mathfrak{s}}$ is independent of the choice of extension $\widehat{p}$.*

3.2. **Kronheimer's invariant and Smirnov's $q-$invariant in higher dimension.** Let (X,ω) be a closed connected oriented symplectic $4-$manifold. We take an almost Kähler $3-$tuple (ω, g, J) on X, where J is compatible to ω and g is determined by ω and J. The symplectic structure ω determines a spin^c structure $\mathfrak{s}_\omega$. Let L_ϵ be a line bundle of X s.t. $c_1(L_\epsilon) = \epsilon$ and let $\mathfrak{s}_\epsilon$ be $\mathfrak{s}_\omega \otimes L_\epsilon$, its corresponding spinor bundles are $S^+ = L_\epsilon \otimes (\Lambda^{0,0} \oplus \Lambda^{0,2})$ and $S^- = L_\epsilon \otimes \Lambda^{0,1}$. There is a preferred spin connection A_0 on K^*_X determined by $g = \omega(\cdot, J\cdot)$. Following Taubes, we define

$$\eta_\rho = \eta_\rho(\omega, J) := -iF^+_{A_0} - \rho\omega. \tag{3.15}$$

Recall that $\mathcal{S}_\omega$ is the space of symplectic forms isotopic to ω. Let $\mathcal{A}_\omega$ be the space of almost complex structures tamed by some symplectic form in $\mathcal{S}_\omega$.

Definition 3.6. We say that a nonzero class $\epsilon \in H^2(X;\mathbb{Z})$ is $\mathcal{A}_\omega$-noneffective if for any $J \in \mathcal{A}_\omega$, $\mathrm{PD}(\epsilon)$ is not represented by a non-constant J-holomorphic subvariety.

Write $K = c_1(K_X)$ for the canonical class. Let $\epsilon \in H^2(X;\mathbb{Z})$ be a class s.t. ϵ is $\mathcal{A}_\omega$-non effective and

$$\epsilon^2 - K \cdot \epsilon = -2k. \tag{3.16}$$

Let $\{\omega_t | t \in S^{2k-1}\} \subseteq \mathcal{S}_\omega$ be a family of symplectic forms based at ω, and let $\alpha \in \pi_{2k-1}(\mathcal{S}_\omega, \omega)$ be the homotopy class represented by it. Then there is a family of almost Kähler metrics $\{g_t = \omega_t(\cdot, J_t\cdot) | t \in S^{2k-1}\}$. Following the construction in (3.15), there is a smooth family

$$\mathfrak{T}_\rho(\omega_t, J_t) := (g_t = \omega_t(\cdot, J_t\cdot), \eta_\rho(t)), \quad t \in S^{2k-1}$$

of parameters in Γ. Let γ_ρ be the image of $\mathfrak{T}_\rho$, when ρ is large enough, γ_ρ is contained in $\Gamma^*_{\mathfrak{s}_\epsilon}$ since ω_t is g_t-self-dual for each $t \in S^{2k-1}$. And by [Theorem 1.3, [29]], γ_ρ is disjoint from $\Delta_{\mathfrak{s}_\epsilon}$. Also, when ρ is large enough, $Pr(\gamma_\rho)$ is contained in $\mathbf{K}_{-[\omega]}(X)$ since

$$\langle [\omega_t], [\omega] \rangle = [\omega]^2 > 0, \quad \forall t \in S^{2k-1}.$$

Thus, there is a family $D = \{(g_b, \eta_b) | b \in \mathbb{D}^{2k}\} \subset \Gamma^*_{\mathfrak{s}_\epsilon}$ extending γ_ρ and $Pr(D) \subset \mathbf{K}_{-[\omega]}$. Following [28], we call D an admissible extension. When D is perturbed to a generic parameter family, the moduli space $\bigcup_{b\in\mathbb{D}^{2k}} \mathcal{M}^{\mathfrak{s}_\epsilon}_{(g_b,\eta_b)}$ is a smooth 0-dimensional smooth manifold.

Definition 3.7. Kronheimer's invariant is defined as

$$Q_\epsilon(\alpha) := \#\left(\bigcup_{b\in\mathbb{D}^{2k}} \mathcal{M}^{\mathfrak{s}_\epsilon}_{g_b,\eta_b}\right) \pmod 2.$$

Lemma 3.8. *Q_ϵ is a homomorphism from $\pi_{2k-1}(\mathcal{S}_\omega, \omega)$ to $\mathbb{Z}_2$.*

Proof. By [13], the family Seiberg-Witten evaluation of $\bigcup_{b\in\mathbb{D}^{2k}} \mathcal{M}^{\mathfrak{s}_\epsilon}_{(g_b,\eta_b)}$ does not depend on the choice of representatives of α, large numbers ρ and smooth family $\{J_t | t \in S^{2k-1}\}$. And it's independent of admissible extensions by Lemma 3.5.

To verify additivity, let $\alpha_1, \alpha_2 \in \pi_{2k-1}(\mathcal{S}_\omega, \omega)$. We choose $\{\omega_t | t \in S^{2k-1}\}$, $\{\omega'_t | t \in S^{2k-1}\}$ in $(\mathcal{S}_\omega, \omega)$, which represent α_1 and α_2 respectively. We take almost complex families $\{J_t | t \in S^{2k-1}\}$, $\{J'_t | t \in S^{2k-1}\}$ which are fiberwise compatible to $\{\omega_t | t \in S^{2k-1}\}$, $\{\omega'_t | t \in S^{2k-1}\}$ in $(\mathcal{S}_\omega, \omega)$. The representative of $\alpha_1 + \alpha_2$ is represented by

$$S^{2k-1} \longrightarrow S^{2k-1} \vee S^{2k-1} \to \{\omega_t | t \in S^{2k-1}\} \vee \{\omega'_t | t \in S^{2k-1}\}$$

and $Q_\epsilon(\alpha_1 + \alpha_2)$ is equal to the Seiberg-Witten invariant on an admissible extension of

$$\{\mathfrak{T}_\rho(\omega_t, J_t) | t \in S^{2k-1}\} \bigcup \{\mathfrak{T}_\rho(\omega'_t, J'_t) | t \in S^{2k-1}\}.$$

Let $D, D' \subset \Gamma^*_{\mathfrak{s}_\epsilon}$ be admissible extension of $\{\mathfrak{T}_\rho(\omega_t, J_t) | t \in S^{2k-1}\}$ and $\{\mathfrak{T}_\rho(\omega'_t, J'_t) | t \in S^{2k-1}\}$, then their union is the desired admissible extension. Hence

$$Q_\epsilon(\alpha_1 + \alpha_2) = Q_\epsilon(\alpha_1) + Q_\epsilon(\alpha_2) \pmod 2.$$

□

3.2.1. *Index sets of Kronheimer invariants on $X\#n\overline{\mathbb{CP}^2}$.* Now let X be T^4 or a $K3$ surface. With a specified non-resonant positive class κ, we define the following index sets

$$\Delta_{T^4} = \{\delta \in \Lambda | \delta \text{ primitive}, \ \delta^2 = 0, \ \kappa \cdot \delta > 0\}, \tag{3.17}$$

$$\Delta_{K3} = \{\delta \in \Lambda | \delta^2 = -2, \ \kappa \cdot \delta > 0\}, \tag{3.18}$$

$$\Delta_{X,k} = \{\delta \in \Delta_X | (k-1)\lambda < \kappa \cdot \delta < k\lambda\}. \tag{3.19}$$

We write Δ and Δ_k when the surface is understood. We say κ is non-resonant if $\kappa \cdot a \neq 0$ for every $0 \neq a \in \Lambda$.

Lemma 3.9. *$\Delta_{X,k}$ is infinite for every $1 \leqslant k \leqslant n$.*

Proof. The proof is essentially the same as [Lemma 10, [28]]. For $\Lambda = 3H$, choose generators f_1, f_2, f_3 of a primitive totally isotropic rank-three sublattice. All classes $\delta_{p,q} := f_1 + pf_2 + qf_3$, $p, q \in \mathbb{Z}$, are primitive and $\delta_{p,q}^2 = 0$. Non-resonance implies that $\kappa \cdot f_2$ and $\kappa \cdot f_3$ are linearly independent over $\mathbb{Q}$. Thus $\{\kappa \cdot (f_1 + pf_2 + qf_3)\}$ is a dense subset of $\mathbb{R}$, and $\Delta_{X,k}$ is infinite. This proves the case when $X = T^4$.

Now let X be a K3 surface and let $\Lambda = 2(-E_8) \oplus 3H$. Let r be a root of a copy of $(-E_8)$ in Λ, i.e. $r^2 = -2$, and choose primitive f_1, f_2 from two distinct copies of H such that $f_1^2 = f_2^2 = 0$.

For every $p, q \in \mathbb{Z}$, the class $\delta_{p,q} = r + pf_1 + qf_2$ satisfies that

$$\delta_{p,q}^2 = r^2 + p^2 f_1^2 + q^2 f_2^2 = -2.$$

By the non-resonance assumption of κ, $\kappa \cdot f_1$ and $\kappa \cdot f_2$ are linearly independent vectors over $\mathbb{Q}$, and $\{\langle \kappa, \delta_{p,q} \rangle | p, q \in \mathbb{Z}\}$ is dense in $\mathbb{R}$. The infiniteness of $\Delta_{X,k}$ follows. □

We set

$$I_k = \{1, ..., k\}. \tag{3.20}$$

Now let $(\tilde{X}, \tilde{\omega})$ be $n-$point blow-up of (X, ω). We assume that the area of each exceptional divisor is equal to λ and $n\lambda$ is small enough. Since $K_{\tilde{\omega}} = e_1 + ... + e_n$, and

$$K_{\tilde{\omega}} - (\delta - \sum_{i \in I_k} e_i) = 2 \sum_{i \in I_k} e_i + \sum_{i \in \{1,...,n\} \setminus I_k} e_i - \delta,$$

the conjugate of $\mathfrak{s}_{\delta - \sum_{i \in I_k} e_i}$ is isomorphic to $\mathfrak{s}_{2\sum_{i \in I_k} e_i + \sum_{i \in \{1,...,n\} \setminus I_k} e_i - \delta}$.

Lemma 3.10. *For each $\delta \in \Delta_{X,k}$, both $\delta - \sum_{i \in I_k} e_i$ and $2\sum_{i \in I_k} e_i + \sum_{i \in \{1,...,n\} \setminus I_k} e_i - \delta$ are $\mathcal{A}_{\tilde{\omega}}$-non-effective.*

Proof. By the definition of $\Delta_{X,k}$, we have $\langle \tilde{\omega}, \delta - \sum_{i \in I_k} e_i \rangle < 0$, thus $\delta - \sum_{i \in I_k} e_i$ is $\mathcal{A}_{\tilde{\omega}}$-non-effective. Since the area of e_i is equal and thus minimal for each $i = 1, ..., n$, E_i is represented by an embedded J-holomorphic curve for each $J \in \mathcal{A}_{\tilde{\omega}}$, see [5]. For each $t = 1, ..., n$, we have

$$e_t \cdot (2 \sum_{i \in I_k} e_i + \sum_{i \in \{1,...,n\} \setminus I_k} e_i - \delta) < 0,$$

If there is $J \in \mathcal{A}_{\tilde{\omega}}$ and a J-holomorphic sub-variety in $\mathrm{PD}(2\sum_{i \in I_k} e_i + \sum_{i \in \{1,...,n\} \setminus I_k} e_i - \delta)$, it consists of a curve in $\mathrm{PD}(\sum_{i \in I_k} e_i - \delta)$ and a component in E_i for each $i = 1, ..., n$.

Applying this strategy to the curve in $\mathrm{PD}(\sum_{i\in I_k} e_i - \delta)$ one more time, we get a curve in $-\mathrm{PD}(\delta)$, which is impossible. □

For each $\delta \in \Delta_{T^4,k}$,

$$d(\mathfrak{s}_{\delta-\sum_{i\in I_k} e_i}) = d(\mathfrak{s}_{K_{\tilde{\omega}}-(\delta-\sum_{i\in I_k} e_i)}) = (\delta - \sum_{i\in I_k} e_i)^2 - \langle \delta - \sum_{i\in I_k} e_i, \sum_{i=1}^{n} e_i\rangle = -2k, \quad (3.21)$$

so the Kronheimer's invariants $Q_{\delta-\sum_{i\in I_k} e_i}$ and $Q_{K_{\tilde{\omega}}-(\delta-\sum_{i\in I_k} e_i)}$ are well-defined homomorphisms from $\pi_{2k-1}(\mathcal{S}_{\tilde{\omega}}, \tilde{\omega})$ to $\mathbb{Z}_2$. We define

$$q_{\delta-\sum_{i\in I_k} e_i} = Q_{\delta-\sum_{i\in I_k} e_i} - Q_{K_{\tilde{\omega}}-(\delta-\sum_{i\in I_k} e_i)} : \pi_{2k-1}(\mathcal{S}_{\tilde{\omega}}, \tilde{\omega}) \to \mathbb{Z}_2.$$

For each $\delta \in \Delta_{K3,k}$,

$$d(\mathfrak{s}_{\delta-\sum_{i\in I_k} e_i}) = d(\mathfrak{s}_{K_{\tilde{\omega}}-(\delta-\sum_{i\in I_k} e_i)}) = (\delta - \sum_{i\in I_k} e_i)^2 - \langle \delta - \sum_{i\in I_k} e_i, \sum_{i=1}^{n} e_i\rangle = -2k-2, \quad (3.22)$$

so the Kronheimer's invariants $Q_{\delta-\sum_{i\in I_k} e_i}$ and $Q_{K_{\tilde{\omega}}-(\delta-\sum_{i\in I_k} e_i)}$ are well-defined homomorphisms from $\pi_{2k+1}(\mathcal{S}_{\tilde{\omega}}, \tilde{\omega})$ to $\mathbb{Z}_2$. We define

$$q_{\delta-\sum_{i\in I_k} e_i} = Q_{\delta-\sum_{i\in I_k} e_i} - Q_{K_{\tilde{\omega}}-(\delta-\sum_{i\in I_k} e_i)} : \pi_{2k+1}(\mathcal{S}_{\tilde{\omega}}, \tilde{\omega}) \to \mathbb{Z}_2.$$

Now, we'll introduce some properties of $q-$invariant.

Lemma 3.11. *Let $\tilde{X}$ be either $K3\#n\overline{\mathbb{CP}}^2$ or $T^4\#n\overline{\mathbb{CP}}^2$, with $n \geqslant 1$. For $r \geqslant 2$, let $f : S^{2r-1} \to \mathrm{Diff}_0(\tilde{X})$ be a smooth based map and set*

$$\pi_f : E_f = (D^{2r}_+ \times \tilde{X}) \bigsqcup (D^{2r}_- \times \tilde{X})/\sim \longrightarrow S^{2r}, \qquad (t,x)_+ \sim (t, f_t(x))_-.$$

Every spin^c *structure $\mathfrak{s}$ on $\tilde{X}$ extends to a* spin^c *structure $\tilde{\mathfrak{s}}$ on $T_{E_f/S^{2r}}$, and the extension is unique up to isomorphism.*

Proof. Equip $D^{2r}_\pm \times \tilde{X}$ with the product spin^c structure $\mathcal{L}_\pm$ induced by $\mathfrak{s}$. After identifying the vertical tangent bundles on the overlap by $\{df_t | t \in S^{2r-1}\}$, $\mathcal{L}_\pm|_{S^{2r-1}\times\tilde{X}}$ differ by a complex line bundle $\mathcal{L} \to S^{2r-1} \times \tilde{X}$ such that $\mathcal{L}|_{\{t\}\times\tilde{X}}$ is trivial for each $t \in S^{2r-1}$. As $r \geqslant 2$, the restriction induces an isomorphism $H^2(S^{2r-1} \times \tilde{X}; \mathbb{Z}) \xrightarrow{\cong} H^2(\tilde{X}; \mathbb{Z})$. Since $\mathcal{L}|_{\{t\}\times\tilde{X}}$ is trivial, $\mathcal{L}$ is trivial, and the two family spin^c-structures glue to a spin^c structure on $T_{E_f/S^{2r}}$.

Let $j_\pm$ be inclusions $D^{2r}_\pm \times \widetilde{X} \hookrightarrow E_f$, and let $k_\pm$ be inclusions $S^{2r-1} \times \tilde{X} \hookrightarrow D^{2r}_\pm \times \widetilde{X}$. Applying the Mayer–Vietoris sequence to $(E_f, D^{2r}_+ \times \tilde{X}, D^{2r}_- \times \tilde{X}, S^{2r-1} \times \tilde{X})$, we have

$$\begin{aligned} 0 \longrightarrow H^2(E_f;\mathbb{Z}) &\xrightarrow{(j_+^*, j_-^*)} H^2(D^{2r}_+ \times \widetilde{X};\mathbb{Z}) \oplus H^2(D^{2r}_- \times \widetilde{X};\mathbb{Z}) \\ &\xrightarrow{k_+^* - k_-^*} H^2(S^{2r-1} \times \widetilde{X};\mathbb{Z}) \longrightarrow 0. \end{aligned} \quad (3.23)$$

Since $r \geqslant 2$ and each f_t is isotopic to the identity, $k_+^* - k_-^*$ can be identified with

$$H^2(\tilde{X};\mathbb{Z}) \oplus H^2(\tilde{X};\mathbb{Z}) \to H^2(\tilde{X};\mathbb{Z}), \quad (a,b) \to a - b.$$

By exactness of (3.23), (j_+^*, j_-^*) is an isomorphism to the diagonal group

$$\{(a,a) : a \in H^2(\widetilde{X};\mathbb{Z})\}.$$

The projection to the first factor is an isomorphism on the diagonal, so j_+^* is an isomorphism. Let $i_+ : \widetilde{X} \hookrightarrow D_+^{2r} \times \widetilde{X}$ be the inclusion of a fiber, i_+^* is an isomorphism between the cohomology groups. Set $\iota = j_+ \circ i_+$,

$$\iota^* = i_+^* \circ j_+^* : H^2(E_f;\mathbb{Z}) \xrightarrow{\cong} H^2(\widetilde{X};\mathbb{Z}).$$

Let $\widetilde{\mathfrak{s}}$ and $\widetilde{\mathfrak{s}}'$ be two extensions of $\mathfrak{s}$, then $\widetilde{\mathfrak{s}}' \cong \widetilde{\mathfrak{s}} \otimes \mathcal{L}$ for a line bundle $\mathcal{L} \to E_f$ such that $\iota^* c_1(\mathcal{L}) = 0$. The injectivity of ι^* implies $c_1(\mathcal{L}) = 0$ and $\mathcal{L}$ is trivial. □

Let r be k when $X = T^4$ and $r = k+1$ when X is a $K3$ surface.

Lemma 3.12. *For $\delta \in \Delta_{X,k}$ and $r \geqslant 2$, set $\epsilon = \delta - \sum_{i\in I_k} e_i$, the composition*

$$q_\epsilon \circ \mathrm{p}_* : \pi_{2r-1}(\mathrm{Diff}_0(\widetilde{X}), \mathrm{id}) \longrightarrow \pi_{2r-1}(\mathcal{S}_{\widetilde{\omega}}, \widetilde{\omega}) \to \mathbb{Z}_2$$

is trivial.

Proof. We follow the argument of [Lemma 6, [28]]. Let $\{f_t | t \in S^{2r-1}\}$ be a family of diffeomorphisms based at id and set $\tilde{\omega}_t = f_t^* \tilde{\omega}$. The family $\{\tilde{\omega}_t \mid t \in S^{2r-1}\}$ represents α. Let

$$E_f = (D_+^{2r} \times \widetilde{X}) \bigcup (D_-^{2r} \times \widetilde{X}) / \sim \longrightarrow S^{2r}, \qquad (t,x)_+ \sim (t, f_t(x))_-.$$

Set $K = K_{\tilde{\omega}}$ and $\epsilon = \delta - \sum_{i\in I_k} e_i$. Then $K - \epsilon$ is $2\sum_{i\in I_k} e_i + \sum_{i\in\{1,\dots,n\}\setminus I_k} e_i - \delta$. By Lemma 3.11, both $\mathfrak{s}_\epsilon$ and $\mathfrak{s}_{K-\epsilon}$ extends to spin^c structures $\mathcal{L}_\epsilon$, $\mathcal{L}_{K-\epsilon}$ on the vertical tangent bundle of E_f. By the uniqueness of extension, $\overline{\mathcal{L}_\epsilon} \cong \mathcal{L}_{K-\epsilon}$.

Let J be an almost complex structure compatible with $\tilde{\omega}$, let $g = \tilde{\omega}(\cdot, J\cdot)$, and set $J_t = f_t^* J$ and $g_t = f_t^* g$. The corresponding perturbation 2-forms $\eta_\rho(\tilde{\omega}_t, J_t)$ satisfy

$$\eta_\rho(\tilde{\omega}_t, J_t) = f_t^* \eta_\rho(\tilde{\omega}, J), \quad t \in S^{2r-1}.$$

We take an admissible extension $\{(g_b, \eta_b) | b \in D_+^{2r}\}$ of $\{(g_t, \eta_\rho(\tilde{\omega}_t, J_t))\}$, and take a trivial perturbation $\{(g, \eta_\rho(\tilde{\omega}, J))\}$ on D_-^{2r}, they form a family of perturbation $\{(g_z, \eta_z) | z \in S^{2r}\}$ on E_f.

Since S^{2r} is simply connected for each $r \geqslant 2$, the chamber of $\{(g_z, \eta_z) | z \in S^{2r}\}$ is determined by the homotopy class of

$$Pr_{\mathcal{L}_\epsilon} : S^{2r} \to \mathbf{K}(\widetilde{X}), \qquad b \longrightarrow H^+_{g_b}(\eta_b) + 2\pi H^+_{g_b}(c_1(L_b)),$$

which is contained in $\mathbf{K}_{-[\tilde{\omega}]}$ by the admissibility of $\{(g_b, \eta_b) | b \in D_+^{2r}\}$. Since $\mathcal{L}_{K-\epsilon} \cong \overline{\mathcal{L}_\epsilon}$ $det(c_1(\mathcal{L}_{K-\epsilon})) = -det(c_1(\mathcal{L}_\epsilon))$, we have

$$Pr_{\mathcal{L}_{K-\epsilon}}(g_z, -\eta_z) = -Pr_{\mathcal{L}_\epsilon}(g_z, \eta_z) \in \mathbf{K}_{[\tilde{\omega}]},$$

which also determines the canonical chamber. By the charge conjugation symmetry,

$$\begin{aligned} Q_\epsilon(\alpha) &= \mathrm{FSW}_{\mathcal{L}_\epsilon}(E_f) \\ &= \mathrm{FSW}_{\mathcal{L}_\epsilon}(E_f;\{(g_z,\eta_z)\}) \\ &= \mathrm{FSW}_{\mathcal{L}_{K-\epsilon}}(E_f;\{(g_z,-\eta_z)\}) \\ &= \mathrm{FSW}_{\mathcal{L}_{K-\epsilon}}(E_f) \\ &= Q_{K-\epsilon}(\alpha). \end{aligned} \tag{3.24}$$

Consequently, $q_\epsilon(\alpha)=0$. □

Definition 3.13. For $r\geqslant 2$ and $\delta\in\Delta_{X,k}$, we define

$$\begin{aligned} \bar{q}_{\delta-\sum_{i\in I_k}e_i} : \mathrm{Ker}(\pi_{2r-2}(\mathrm{Symp}_{st}(\widetilde{X},\widetilde{\omega}))\to\pi_{2r-2}(\mathrm{Diff}_0(\widetilde{X}))) &\longrightarrow \mathbb{Z}_2, \\ \beta &\longrightarrow q_{\delta-\sum_{i\in I_k}e_i}(\alpha), \end{aligned} \tag{3.25}$$

where $\alpha\in\pi_{2r-1}(\mathcal{S}_{\widetilde{\omega}},\widetilde{\omega})$ satisfies $\partial_*\alpha=\beta$.

3.3. **Family blowups and Seiberg–Witten equations on Kähler surfaces.** Let $\pi:\mathcal{X}\to(B,0)$ be a proper holomorphic submersion with compact connected surface fibers, where $B\subset\mathbb{C}^n$ is a polydisk. Write $X_b=\pi^{-1}(b)$ and $X=X_0$. In this section, we relate the family Seiberg-Witten equation on $\mathcal{X}$, to the family Seiberg-Witten equation of a $D^2\times B$ family of $\tilde{X}$.

Choose a smooth trivialization over B, we can identify $\mathcal{X}$ with a smooth family $\{J_b|b\in B\}$ of complex structures, and let $\{(\omega_b,J_b)|b\in B\}$ be a smooth family of Kähler structures. Fix a smooth complex line bundle $\mathcal{L}_\alpha$ on $\mathcal{X}$ with fiberwise first Chern class α. As before, $\mathfrak{s}_\alpha$ denotes the spin^c structure determined by $\mathfrak{s}_{\omega_b}\otimes\mathcal{L}_\alpha|_{X_b}$. Thus

$$c_1(\det\mathfrak{s}_\alpha)=2\alpha-K_{X_b},\qquad d_X(\mathfrak{s}_\alpha)=\alpha^2-K_X\cdot\alpha.$$

We use the Kähler metrics $g_b=\omega_b(\cdot,J_b\cdot)$ and Taubes' perturbations $\eta_\rho(\omega_b,J_b)$ introduced above. When ρ is large enough for the family, $\mathfrak{T}_\rho(\omega_b,J_b)=(g_b,\eta_\rho(\omega_b,J_b))$ will be contained in $\Gamma^*_{\mathfrak{s}_\alpha}$.

For a smooth curve $C\subset X$ such that $[C]=\mathrm{PD}(\alpha)$, let Θ_X be the sheaf of holomorphic vector fields and write

$$N=\mathcal{N}_{C|X}=T_X|_C/T_C,\qquad r_*:H^1(X,\Theta_X)\longrightarrow H^1(C,N)$$

for normal restriction. Let $\mathrm{KS}:T_0B\to H^1(X,\Theta_X)$ be the Kodaira–Spencer map. By [30], a smooth curve $C\subset X_0$ represents a solution of $\bigcup_{b\in B}\mathcal{M}^{\mathfrak{s}_\alpha}_{\eta_\rho(\omega_b,J_b)}$ when ρ is big enough. And by [8][13][26], the solution corresponding to C is regular if

$$r_*\circ\mathrm{KS}:T_0B\longrightarrow H^1(C,N) \tag{3.26}$$

is surjective.

Assume that C is the unique effective divisor in class α among all fibers of $\mathcal{X}$. Assume that

$$H^0(C,N)=0,\qquad d_X(\mathfrak{s}_\alpha)+2n=0, \tag{3.27}$$

and that (3.26) is surjective.

Let $\Delta \subset \mathbb{C}$ be a disk centred at 0, and let $F : B \times \Delta \to \mathcal{X}$ be a jointly holomorphic map such that $\pi \circ F = \mathrm{pr}_B$ and $F(0,0) = p \in C$. Assume that

$$\nu : T_0\Delta \longrightarrow N|_p, \qquad \nu(w) = d(F|_{\{0\}\times\Delta})_0(w) \bmod T_pC \tag{3.28}$$

is nonzero, and hence an isomorphism of complex lines.

Set a holomorphic family $\mathcal{X}_\Delta$ and a section σ to it

$$\mathcal{X}_\Delta = (B \times \Delta) \times_B \mathcal{X}, \qquad \pi_\Delta : \mathcal{X}_\Delta \longrightarrow B \times \Delta, \qquad \sigma(b,z) = ((b,z), F(b,z)).$$

Blowing up along the image of σ gives

$$\Pi : \widetilde{\mathcal{X}} = \mathrm{Bl}_{\sigma(B\times\Delta)}\, \mathcal{X}_\Delta \longrightarrow \mathcal{X}_\Delta, \quad \widetilde{\Pi} = \pi_\Delta \circ \Pi : \widetilde{\mathcal{X}} \to B \times \Delta.$$

A fiber of the $\tilde{\Pi}$ is $\widetilde{X}_{b,z} = \mathrm{Bl}_{F(b,z)} X_b$, and the restriction $\pi_{b,z} = \Pi|_{\widetilde{X}_{b,z}} : \widetilde{X}_{b,z} \to X_b$ is the fiberwise blowdown. Let E be the relative exceptional divisor and e its fiberwise cohomology class.

Set the line bundle $\widetilde{\mathcal{L}}_\alpha = \Pi^* \mathrm{pr}_{\mathcal{X}}^* \mathcal{L}_\alpha \otimes \mathcal{O}_{\widetilde{\mathcal{X}}}(-E)$, its fiberwise first Chern class is $\alpha - e$. Shrinking B and Δ if necessary, there is a family $\{\widetilde{\omega}_{b,z}\}$ of Kähler forms of class $\pi_{b,z}^*[\omega_b] - \varepsilon e$ for a fixed sufficiently small $\varepsilon > 0$.

Theorem 3.14. *For all sufficiently large ρ, the family moduli space*

$$\bigcup_{(b,z)\in B\times\Delta} \mathcal{M}^{\mathfrak{s}_{\alpha-e}}_{\eta_\rho(\widetilde{\omega}_{b,z}, \widetilde{J}_{b,z})}$$

consists of a unique solution which is regular, represented by the strict transform $\widetilde{C} \subset \widetilde{X}_{0,0}$. In other word, the restriction of the Kodaira–Spencer map at $(0,0)$ is surjective onto $H^1(\widetilde{C}, \mathcal{N}_{\widetilde{C}|\widetilde{X}_{0,0}})$.

Let $\pi_0 : \widetilde{X} \to X$ be the blowdown at $p \in C$. Identify the strict transform $\widetilde{C}$ with C by $f := \pi_0|_{\tilde{C}}$, and write $\mathfrak{m}_p$ for the ideal sheaf of p. and $d\pi_0$ identifies

$$\mathcal{N}_{\widetilde{C}|\widetilde{X}} \cong f^*(N(-p)), \qquad N(-p) = N \otimes \mathcal{O}_C(-p). \tag{3.29}$$

Lemma 3.15. *With the identifications above, there are exact sequences*

$$H^0(X, \Theta_X) \xrightarrow{\mathrm{ev}_p} T_pX \xrightarrow{\delta_p} H^1(\widetilde{X}, \Theta_{\widetilde{X}}) \xrightarrow{\beta} H^1(X, \Theta_X) \longrightarrow 0 \tag{3.30}$$

and

$$H^0(C, N) \xrightarrow{\mathrm{ev}_p} N|_p \xrightarrow{\partial_N} H^1(C, N(-p)) \xrightarrow{\gamma} H^1(C, N) \longrightarrow 0. \tag{3.31}$$

Let $r_* : H^1(X, \Theta_X) \to H^1(C, N)$ and $\widetilde{r}_* : H^1(\tilde{X}, \Theta_{\tilde{X}}) \to H^1(\tilde{C}, N_{\tilde{C}|\tilde{X}})$ be restriction maps, $\mathrm{pr}_N : T_pX \to N|_p = T_pX/T_pC$ is the quotient map. Then

$$\gamma \circ \widetilde{r}_* = r_* \circ \beta, \qquad \widetilde{r}_* \circ \delta_p = \partial_N \circ \mathrm{pr}_N, \tag{3.32}$$

In particular,

$$\widetilde{r}_*(\ker \beta) = \ker \gamma. \tag{3.33}$$

Proof. For one point blowup in the complex-analytic category, the standard tangent-sheaf identities are

$$(\pi_0)_*\Theta_{\widetilde{X}} = \mathfrak{m}_p\Theta_X, \qquad R^1(\pi_0)_*\Theta_{\widetilde{X}} = 0; \tag{3.34}$$

see the Proof of [Proposition 2.6[23]], where these identities are used for compact complex manifolds. The first identification is induced by $d\pi_0$. By the five-term exact sequence of the Leray spectral sequence, we have

$$0 \longrightarrow E_2^{1,0} = H^1(X, (\pi_0)_*\Theta_{\widetilde{X}}) \longrightarrow H^1(\widetilde{X}, \Theta_{\widetilde{X}}) \longrightarrow E_2^{0,1} = H^0(X, R^1(\pi_0)_*\Theta_{\widetilde{X}}) \cong \{0\}$$
$$\xrightarrow{d_2} E_2^{2,0} \longrightarrow H^2(\widetilde{X}, \Theta_{\widetilde{X}}),$$

thus $H^1(X, \mathfrak{m}_p\Theta_X) \cong H^1(\widetilde{X}, \Theta_{\widetilde{X}})$. Under this identification, (3.30) is induced by the evaluation sequence

$$0 \to \mathfrak{m}_p\Theta_X \to \Theta_X \to (T_pX)_p \to 0,$$

where $(V)_p$ denotes the skyscraper sheaf with fiber V at p. This defines δ_p and β.

Choose a coordinate system (x, y) near p with $C = \{x = 0\}$. Let q be the unique point of $\widetilde{C} \cap \pi_0^{-1}(p)$. There are holomorphic coordinates (u, y) on a neighbourhood of q in which

$$\pi_0(u, y) = (uy, y), \qquad \widetilde{C} = \{u = 0\}.$$

In these coordinates, the restriction $f = \pi_0|_{\widetilde{C}}$, $(0, y) \mapsto (0, y)$ is a biholomorphism from $\widetilde{C}$ to C. $d\pi_0$ induces a morphism from $N_{\widetilde{C}|\widetilde{X}}$ to N, and its local expression is$[\partial_u] \longmapsto y[\partial_x]$. This identifies $N_{\widetilde{C}/\widetilde{X}}$ with $N(-p)$, and the morphism induced by $d\pi_0$ is identified with natural inclusion $N(-p) \hookrightarrow N$. The long exact sequence corresponding to

$$0 \longrightarrow N(-p) \longrightarrow N \xrightarrow{\mathrm{ev}_p} (N|_p)_p \longrightarrow 0$$

therefore yields (3.31).

Write $i : C \hookrightarrow X$. Restrictions of the sheaves give the following diagram of sheaves on X with exact rows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathfrak{m}_p\Theta_X & \longrightarrow & \Theta_X & \longrightarrow & (T_pX)_p & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle R} & & \downarrow{\scriptstyle r} & & \downarrow{\scriptstyle \mathrm{pr}_N} & & \\ 0 & \longrightarrow & i_*N(-p) & \longrightarrow & i_*N & \longrightarrow & (N|_p)_p & \longrightarrow & 0. \end{array} \tag{3.35}$$

Here, r is normal restriction to C, and R is its restriction to $\mathfrak{m}_p\Theta_X$, taking values in $i_*N(-p)$. The left square commutes by the identification with $d\pi_0$ just established, and the right square commutes by the property of the evaluation at p. The isomorphism $H^1(X, \mathfrak{m}_p\Theta_X) \cong H^1(\widetilde{X}, \Theta_{\widetilde{X}})$ identifies $H^1(R)$ with $\widetilde{r}_*$, and for sheaves supported on C, the same comparison is the cohomology identification induced by the biholomorphism f. Taking cohomology of (3.35) and using naturality of connecting homomorphisms, we get (3.32). Finally, by the exactness and the surjectivity of pr_N, we have

$$\widetilde{r}_*(\ker\beta) = \widetilde{r}_*(\operatorname{im}\delta_p) = \operatorname{im}(\partial_N \mathrm{pr}_N) = \operatorname{im}\partial_N = \ker\gamma.$$

□

Choose an open coordinate neighbourhood $U_0 \subset X$ of p, with coordinates $z = (z_1, z_2)$ satisfying $z(p) = 0$. Let $U \subset U_0$ be an open neighbourhood of p and set $\mathcal{Y} := \{\mathrm{Bl}_q X | q \in U\}$. Let v be a vector in T_pU, we can regard v as a global section of the skyscraper sheaf $(T_pX)_p$. Then $\delta_p(v)$ is its image in the connecting homomorphism of the long exact sequence

$$H^0(X, \mathfrak{m}_p\Theta_X) \longrightarrow H^0(X, \Theta_X) \to H^0(X, (T_pX)_p) \to H^1(X, \mathfrak{m}_p\Theta_X) \to ...$$

By the proof of [Theorem 4.2, [22]], the Kodaira–Spencer map $\mathrm{KS}_p(v)$ of $\mathcal{Y}$ at p is its image under the connecting homomorphism associated with

$$0 \to H^0(X, (\pi_0)_*\Theta_{\tilde{X}}) \to H^0(X, \Theta_X) \to H^0(X, (T_pX)_p) \to H^1(X, (\pi_0)_*\Theta_{\tilde{X}}) \to ...$$

Under the identifications in Lemma 3.15, KS_p is identified with δ_p. Consequently, we have

$$\mathrm{Im}\,\mathrm{KS}_p = \ker\beta, \qquad \ker\mathrm{KS}_p = \mathrm{Im}\big(H^0(X, \Theta_X) \xrightarrow{\mathrm{ev}_p} T_pX\big).$$

And now we are ready to prove the induction of transversality of solution of family Seiberg–Witten equations.

Proof of Theorem 3.14. Set $F_0(z) = F(0, z)$. By (3.28), after shrinking Δ, we have $F_0(\Delta) \subset U$ and $F_0^{-1}(C) = \{0\}$. For $(b, z) \in B \times \Delta$, write

$$\pi_{b,z} : \widetilde{X}_{b,z} = \mathrm{Bl}_{F(b,z)} X_b \longrightarrow X_b$$

let $\tilde{\mathcal{X}}$ be $\{\mathrm{Bl}_{F(b,z)} X_b | b \in B, z \in \Delta\}$ and let $\{\tilde{\omega}_{b,z} | (b, z) \in B \times \Delta\}$ be a smooth family of Kähler forms on $\tilde{\mathcal{X}}$. For sufficiently large ρ,elements in $\bigcup_{(b,z)\in B\times\Delta} \mathcal{M}^{\mathfrak{s}_{\alpha-e}}_{\eta_\rho(\tilde{\omega}_{b,z}, \tilde{J}_{b,z})}$ are in one-to-one correspondence with effective divisors in $\alpha - e$. The blowdown of any such divisor is an effective divisor in class α on some X_b. By the uniqueness hypothesis, it must be $C \subset X_0$. Since $F_0^{-1}(C) = \{0\}$ and C is smooth, its proper transform $\widetilde{C} \subset \widetilde{X}_{0,0}$ is the unique effective divisor in the class. Hence the family moduli space consists of exactly one solution.

It remains to prove regularity of the point in $\bigcup_{(b,z)\in B\times\Delta} \mathcal{M}^{\mathfrak{s}_{\alpha-e}}_{\eta_\rho(\tilde{\omega}_{b,z}, \tilde{J}_{b,z})}$. Write

$$\mathrm{KS} : T_0B \longrightarrow H^1(X, \Theta_X), \qquad \mathrm{KS}' : T_0B \oplus T_0\Delta \longrightarrow H^1(\widetilde{X}, \Theta_{\tilde{X}}),$$

for the Kodaira-Spencer map of $\mathcal{X}$ and $\tilde{\mathcal{X}}$, and set

$$L = \widetilde{r}_* \circ \mathrm{KS}' : T_0B \oplus T_0\Delta \longrightarrow H^1(C, N(-p)).$$

By (3.31) and that $H^0(C, N) = 0$, we have

$$0 \longrightarrow N|_p \xrightarrow{\partial_N} H^1(C, N(-p)) \xrightarrow{\gamma} H^1(C, N) \longrightarrow 0. \tag{3.36}$$

Since $\{\pi_{b,z}(\tilde{X}_{b,z}) \mid (b, z) \in B \times \Delta\}$ can be identified with the pullback family $\mathrm{pr}_B^*\,\mathcal{X} \to B \times \Delta$, where pr_B is the projection $B \times \Delta \to B$. By naturality of the Kodaira–Spencer map, $\beta \circ \mathrm{KS}'(v, w) = \mathrm{KS}(v)$. Together with Lemma 3.15, this implies

$$\gamma \circ L(v, w) = r_* \circ \mathrm{KS}(v).$$

Thus $\gamma \circ L$ is surjective, and $L(\{0\} \oplus T_0\Delta) \subseteq \ker\gamma$.

The family $\{\tilde{X}_{0,z}|z\in\Delta\}$ is the pullback of $\mathcal{Y}\to U$ by F_0, so

$$\mathrm{KS}'(0,w)=\delta_p\big(dF_0(w)\big).$$

Using $\widetilde{r}_*\circ\delta_p=\partial_N\circ\mathrm{pr}_N$ from Lemma 3.15, we therefore obtain

$$\widetilde{r}_*\mathrm{KS}'(0,w)=\partial_N\big(\nu(w)\big)\in\ker\gamma, \tag{3.37}$$

where $\nu=\mathrm{pr}_N\circ dF_0: T_0\Delta\to N|_p$. Since $dim(\ker\gamma)=1$ and $\partial_N\big(\nu(w)\big)\neq 0$ for $w\neq 0$, we have $L(\{0\}\oplus T_0\Delta)=\ker\gamma$. By (3.36), L is surjective, and the solution in $\bigcup_{(b,z)\in B\times\Delta}\mathcal{M}^{\mathfrak{s}_{\alpha-e}}_{\eta_\rho(\widetilde{\omega}_{b,z},\widetilde{J}_{b,z})}$ represented by $\widetilde{C}$ is regular. □

4. Local linking families and the proof of the main theorem

Let X be a $K3$ surface or T^4, and assume that κ is non-resonant. Throughout this section, take $1\leqslant k\leqslant n$ for K3 surfaces and $2\leqslant k\leqslant n$ for tori, so that $r\geqslant 2$. For a positive class κ, denote $\mathcal{X}_\kappa\to\mathcal{M}_\kappa$ the fine moduli space of K3 surfaces and T^4, and denote X_u by the complex surface with the marking determined by $u\in\mathcal{M}_\kappa$. For $q=(u,\mathbf{x})\in\mathcal{P}_\kappa$, write $\tilde{X}_q=\mathrm{Bl}_{\mathbf{x}}X_u$. Take $\lambda<\frac{\sqrt{\kappa^2}}{2n}$, set $\boldsymbol{\lambda}=(\lambda,\ldots,\lambda)$ and $S=n\lambda$, and put

$$\alpha_{\kappa,\boldsymbol{\lambda}}=\kappa-\lambda\sum_{i=1}^n e_i,\qquad K=\sum_{i=1}^n e_i,\qquad I_k=\{1,\ldots,k\}.$$

Recall the index set $\Delta_{X,k}$ in Section 3.2, for each $\delta\in\Delta_{X,k}$,

$$(k-1)\lambda<\kappa\cdot\delta<k\lambda,\qquad \alpha_{\kappa,\boldsymbol{\lambda}}\cdot(\delta-\sum_{i\in I_k}e_i)<0. \tag{4.1}$$

4.1. **Local incidence loci and transverse disks.** Let $\delta\in\Delta_{X,k}$, Since $\delta^2\leqslant 0$ with $\kappa\cdot\delta\neq 0$, the orthogonal projection $v\in\kappa^\perp\subset\Lambda_{\mathbb{R}}$ of δ satisfies $v^2<0$. By the properties of lattice $\Lambda_{\mathbb{R}}$, we can see that $H^\kappa_\delta:=H_\delta\bigcap\mathcal{M}_\kappa\subset\mathcal{M}_X$ is a nonempty hypersurface of $\mathcal{M}_\kappa$. Let u_0 be a generic point of it, i.e. $\mathrm{NS}(X_{u_0})=\mathbb{Z}\delta$.

Lemma 4.1. *There is a smooth connected holomorphic curve $C_0\subset X_{u_0}$ representing δ.*

Proof. Recall that $\mathrm{NS}(X_{u_0})=\mathbb{Z}\delta$ and that κ is a Kähler class with $\kappa\cdot\delta>0$.

Suppose first that X_{u_0} is a K3 surface. Since $\delta^2=-2$, either δ or $-\delta$ is effective; see [§3, Chapter VIII, [4]]. The latter does not happen since $\kappa\cdot\delta>0$. Since $\mathrm{NS}(X_{u_0})=\mathbb{Z}\delta$ and δ is primitive, any holomorphic curve C representing δ must be reduced and irreducible. By the adjunction formula, $p_a(C)=1+\delta^2/2=0$, $C_0:=C$ is a smooth rational curve.

Suppose now that X_{u_0} is a complex torus T^4. Since δ is a primitive isotropic integral class of type $(1,1)$, by the construction in the proof of [Lemma 11, [28]], there is a regular fibration

$$\pi: X_{u_0}\longrightarrow E, \tag{4.2}$$

over an elliptic curve E such that each fiber of it is a connected smooth elliptic curve representing $\pm\delta$. Since $\kappa\cdot\delta>0$, there is a smooth curve C_0 in δ. □

Recall the bound (2.6). By Lemma 2.8, hyperplanes H_a^κ such that

$$-c_X \leqslant a^2 \leqslant 0, \quad 0 < \kappa \cdot a \leqslant (n + \sqrt{c_X n})\lambda, \tag{4.3}$$

are locally finite. Here, $c_X = 2$ for $K3$ surfaces and $c_X = 0$ for T^4. Thus, we can take a holomorphic family $\{X_u | u \in U\}$ such that the base U is a neighborhood of u_0, with $U \bigcap H_a^\kappa \neq \varnothing$ if and only if $H_a = H_\delta$, i.e. a is equal to (a multiple of) δ. Take a holomorphic disk $D_\delta \subset U$ with coordinate $z \in \mathbb{D}^2$, intersecting H_δ^κ once transversally at u_0, where $z = 0$.

Recall from Section 2.3 the projection $r : \overline{\mathcal{P}}_\kappa = \mathrm{Conf}_n(\mathcal{X}_\kappa/\mathcal{M}_\kappa) \longrightarrow \mathcal{M}_\kappa$, $\mathcal{P}_\kappa = \overline{\mathcal{P}}_\kappa$ for K3 surfaces and $\mathcal{P}_\kappa = \{x_1 = Z(u)\} \subset \overline{\mathcal{P}}_\kappa$ for tori. We take the pullback family

$$p : \mathcal{P}_{D_\delta} := \mathcal{P}_\kappa \bigcap r^{-1}(D_\delta) \to D_\delta$$

by the inclusion $D_\delta \hookrightarrow U \subset \mathcal{M}_\kappa$. When X is a $K3$ surface, there is a unique smooth rational curve $C_0 \subset X_{u_0}$ of self-intersection -2. When X is a torus, there is a unique smooth elliptic curve C_0 on X_{u_0} of self-intersection 0 and contains $Z(u_0)$. For $q = (u, x_1, \ldots, x_n) \in \mathcal{P}_{D_\delta}$, we set $\tilde{X}_q$ to be the blowup of X_u along $x_1, \cdots, x_n$. By the definition in (2.14), the fibers of $r|_{\mathcal{P}_\kappa}$ are of complex dimension $2n$ and $2n-2$ for $K3$ surface and T^4, respectively. Thus

$$\dim_{\mathbb{C}} \mathcal{P}_{D_\delta} = \begin{cases} 2n+1, & X \text{ a K3 surface}, \\ 2n-1, & X = T^4. \end{cases}$$

Define

$$\mathcal{H}_\delta := \big\{(u_0, y_1, \ldots, y_n) \in \mathcal{P}_{D_\delta} | y_i \in C_0 \text{ for every } i \in I_k\big\}.$$

$\mathcal{H}_\delta$ is the incidence locus for complex surfaces in $\tilde{X}_q$ such that there is a holomorphic curve in $\delta - \sum_{i \in I_k} e_i$. It is a holomorphic sub-manifold of $\mathcal{P}_{D_\delta}$.

We say a point $q_0 = (u_0, x_1^0, \ldots, x_n^0) \in \mathcal{H}_\delta$ is generic if

$$x_i^0 \in C_0 \quad (i \in I_k), \qquad x_j^0 \notin C_0 \quad (j \notin I_k),$$

and when $X = T^4$, we further require that the points x_j^0, $j \notin I_k$, lie on pairwise distinct fibers of the elliptic fibration associated with δ, none of which is C_0.

For K3 surfaces, $\mathcal{H}_\delta$ can be considered as an open subset of $C_0^k \times X_{u_0}^{n-k}$ consisting of pairwise distinct points. For $X = T^4$, it is an open subset of $C_0^{k-1} \times X_{u_0}^{n-k}$ consisting of pairwise distinct points different from $Z(u_0)$. As a result, the dimension $\mathcal{H}_\delta$ is

$$\dim_{\mathbb{C}} \mathcal{H}_\delta = \begin{cases} k + 2(n-k) = 2n - k, & X \text{ a K3 surface}, \\ (k-1) + 2(n-k) = 2n - k - 1, & X = T^4. \end{cases}$$

Consequently,

$$r := \mathrm{codim}_{\mathbb{C}}(\mathcal{H}_\delta, \mathcal{P}_{D_\delta}) = \dim_{\mathbb{C}} \mathcal{P}_{D_\delta} - \dim_{\mathbb{C}} \mathcal{H}_\delta = \begin{cases} k+1, & X \text{ a K3 surface}, \\ k, & X = T^4. \end{cases}$$

Let B_δ be the image of a sufficiently small holomorphic ball embedding $i : \mathbb{D}^{2r} \hookrightarrow \mathcal{P}_{D_\delta}$ such that $\{q_0\} = B_\delta \bigcap \mathcal{H}_\delta$, q_0 is a generic point, and B_δ intersects $\mathcal{H}_\delta$ transversally. Suppose

that the intersection point $q_0 = (u_0, x_1, ..., x_n)$ is a generic point of $\mathcal{H}_\delta$. Pull back the blowup family (2.16) by i, and denote it by

$$p_\delta : \widetilde{\mathcal{X}}_\delta \to B_\delta \tag{4.4}$$

with blowdown maps $\pi_q : \widetilde{X}_q \to X_{u(q)}$.

4.2. **Kähler structures on B_δ with described boundaries.** In this section, we will use the parameter families S_δ to construct elements in $\pi_{2r-1}(\mathcal{S}_{\tilde{\omega}}, \tilde{\omega})$.

Lemma 4.2. *The punctured ball $B_\delta \backslash \{q_0\}$ lies in $\mathcal{P}_{\kappa,\boldsymbol{\lambda}}$. In particular, $S_\delta = \partial B_\delta$ lies in $\mathcal{P}_{\kappa,\boldsymbol{\lambda}}$.*

Proof. By Lemma 2.2, if $\alpha_{\kappa,\boldsymbol{\lambda}}$ is not a Kähler class of $\tilde{X}_q$ for some $q = (u, x_1, \ldots, x_n) \in B_\delta$, then there is a reduced irreducible curve $C \subset X_u$ such that $\alpha_{\kappa,\boldsymbol{\lambda}} \cdot [\widetilde{C}] \leqslant 0$ and

$$-c_X \leqslant [C]^2 \leqslant 0, \qquad 0 < \kappa \cdot [C] \leqslant (n + \sqrt{c_X n})\lambda.$$

By the local finiteness in Lemma 2.8 and the choice of D_δ in Section 4.1, we may assume that

(1) $D_\delta \cap H_a = \varnothing$ for each $a \in \mathcal{I}_X(S)$ such that $H_a \neq H_\delta$,
(2) $D_\delta \bigcap H_\delta = \{u_0\}$,
(3) for each $i \notin I_k$ and each $(u_0, x_1, ..., x_n) \in B_\delta$, $x_i \notin C_0$.

If $u \neq u_0$, then the choice of D_δ implies that $u \notin H_a$ for each $a \in \mathcal{I}_X(S)$. Then by Lemma 2.7 and Lemma 2.8, $u \in \mathcal{G}_{\kappa,n\lambda} \subseteq \mathcal{M}_{\kappa,n\lambda} \subset \mathcal{M}_{\kappa,\boldsymbol{\lambda}}$, and $\alpha_{\kappa,\boldsymbol{\lambda}}$ is a Kähler class for any $q = (u, x_1, ..., x_n) \in B_\delta$.

It remains to consider $u = u_0$. We distinguish the two cases.

Suppose first that X_{u_0} is a K3 surface. Since $\mathrm{NS}(X_{u_0}) = \mathbb{Z}\delta$, every reduced irreducible curve $C \subset X_{u_0}$ has class $[C] = m\delta$ for some $m \in \mathbb{Z}$. Since $\kappa \cdot [C] > 0$, we have $m > 0$. By the adjunction formula, $m = 1$ can be deduced from

$$p_a(C) = 1 - m^2 \geqslant 0.$$

Thus C_0 is the unique reduced irreducible curve on X_{u_0}. Let $\tilde{C}_0$ be the proper transform of C_0 in $\tilde{X}_q$ and set

$$m_{C_0}(q) = \#\{i : x_i \in C_0\}.$$

Since C_0 is smooth, the multiplicities at each point of it are equal to 1, thus

$$\alpha_{\kappa,\boldsymbol{\lambda}} \cdot [\widetilde{C}_0] = \kappa \cdot \delta - \lambda m_{C_0}(q).$$

By shrinking B_δ if necessary, $x_i \notin C_0$ for each $i \notin I_k$ and each $(u_0, x_1, ..., x_n) \in B_\delta$. Since $B_\delta \bigcap \mathcal{H}_\delta = \{q_0\}$, $m_{C_0}(q) \leqslant k - 1$ for each $q \neq q_0$. By definition of $\Delta_{K3,k}$,

$$\alpha_{\kappa,\boldsymbol{\lambda}} \cdot [\widetilde{C}_0] \geqslant \kappa \cdot \delta - (k-1)\lambda > 0,$$

and $\alpha_{\kappa,\boldsymbol{\lambda}}$ is Kähler since it satisfies all conditions in Lemma 2.1.

Now suppose that X_{u_0} is a complex two-torus. Since $\mathrm{NS}(X_{u_0}) = \mathbb{Z}\delta$, for every reduced irreducible curve $C \subset X_{u_0}$, $[C] = m\delta$ for some $m > 0$. Let π be the holomorphic elliptic fibration (4.2) with fibers in class δ, the image of C under π is a point since $C \cdot \delta = 0$.

Since C is irreducible and contained in a fiber of π, it is a fiber of π, and consequently $[C] = \delta$.

Recall the definition of C_0 and let $\pi(C_0) = p_0$ and let U_{p_0} be a small neighborhood of p_0. By shrinking B_δ if necessary, $\pi(x_i)$ is contained in U_{p_0} for each $i \in I_k$, whereas $\pi(x_i) \notin U_{p_0}$ for each $i \in \{1, ..., n\} \backslash I_k$, and by our assumption, $\pi(x_i) \neq \pi(x_j)$ for $i \neq j \in \{1, ..., n\} \backslash I_k$. For $q \neq q_0$, no fiber contains all k points in $\{x_1, ..., x_n\}$, since it happens only when $q \in \mathcal{H}_\delta$. Thus

$$\alpha_{\kappa,\boldsymbol{\lambda}} \cdot [\widetilde{C}] = \kappa \cdot \delta - \lambda \#\{i : x_i \in C\} \geqslant \kappa \cdot \delta - (k-1)\lambda > 0.$$

Again, $\alpha_{\kappa,\boldsymbol{\lambda}}$ is Kähler since it satisfies all conditions in Lemma 2.1. This completes the proof of the Lemma. □

Lemma 4.3. *There is a smooth family $\{\tilde{\omega}_q | q \in B_\delta\}$ of Kähler forms such that*

(1) $[\tilde{\omega}_q] = \alpha_{\kappa,\boldsymbol{\lambda}}$ if q is contained in a collar of S_δ.
(2) $\langle [\tilde{\omega}_q], \delta \rangle > 0$ for each $q \in B_\delta$.

Proof. By Lemma 4.2, for each $q \in B_\delta \backslash \{q_0\}$, there is a small neighborhood U_q of q and a smooth family $\{\tilde{\omega}_{q,q'} | q' \in U_q\}$ of symplectic forms such that $\tilde{\omega}_{q,q'}$ is a Kähler form of $\tilde{X}_{q'}$ in the class $\alpha_{\kappa,\boldsymbol{\lambda}}$.

Since δ is represented by an effective divisor in $\tilde{X}_{q_0}$, there is a Kähler $\tilde{\omega}_{q_0}$ on $\tilde{X}_{q_0}$ such that $\langle \tilde{\omega}_{q_0}, \delta \rangle > 0$. And by the Kodaira-Spencer stability theorem [11], there is a small neighborhood U_{q_0} of q_0 such that $\tilde{\omega}_{q_0}$ extends to a smooth family $\{\tilde{\omega}_{q_0,q} | q \in U_{q_0}\}$ of Kähler forms with $\langle \tilde{\omega}_{q_0,q}, \delta \rangle > 0$ for each $q \in U_{q_0}$. Since B_δ is compact, there are finitely many points $\{q_1, ..., q_m\} \subset B_\delta \backslash \{0\}$ such that $\mathcal{U} := \{U_{q_0}, U_{q_1}, ..., U_{q_m}\}$ is an open cover of B_δ.

Choose a smooth partition of unity $\{\chi_0, \chi_1, ..., \chi_m\}$ subordinate to $\mathcal{U}$. By the construction in [§ 3, [15]], define

$$\tilde{\omega}_q = \sum_{i=0}^{m} \chi_i(q) \tilde{\omega}_{q_i,q},$$

then $\tilde{\omega}_q$ is Kähler by convexity of the space of Kähler forms of a compact complex surface. Since U_{q_0} is small, we have $[\tilde{\omega}_q] = \alpha_{\kappa,\boldsymbol{\lambda}}$ for each q in a collar of ∂B_δ. When $q \notin U_{q_0}$, the second condition is satisfied since $\langle \alpha_{\kappa,\boldsymbol{\lambda}}, \delta \rangle > 0$. When $q \in U_{q_0}$, since

$$\langle \alpha_{\kappa,\boldsymbol{\lambda}}, \delta \rangle > 0, \quad \langle \tilde{\omega}_{q_i,q}, \delta \rangle > 0, \quad \chi_i(q) \geqslant 0, \quad \sum_{i=0}^{m} \chi_i(q) \equiv 1,$$

we have

$$\langle \tilde{\omega}_q, \delta \rangle = \sum_{i=0}^{m} \chi_i(q) \langle \tilde{\omega}_{q_i,q}, \delta \rangle > 0$$

and the second condition is satisfied. □

Choose $q_* = (u_*, \mathbf{x}_*) \in S_\delta$ with $u_* \neq u_0$. Since B_δ is contractible, there is a smooth trivialization $\Psi_q : \widetilde{X}_{q_*} \to \widetilde{X}_q$ such that $\Psi_{q_*} = id$. And $\{\Psi_q^* \tilde{\omega}_q | q \in S_\delta\}$ defines an element of $\pi_{2r-1}(\mathcal{S}_{\tilde{\omega}_{q_*}}, \tilde{\omega}_{q_*})$.

Let $(\widetilde{X},\widetilde{\omega})$ be as in Theorem A, we can identify to a Kähler form $\widetilde{\omega}_q$ of $\tilde{X}_q$ where $q\in\mathcal{P}_{\kappa,\boldsymbol{\lambda}}$. Recall that $\mathcal{P}_{\kappa,\boldsymbol{\lambda}}\subset\overline{\mathcal{P}}_{\kappa,\boldsymbol{\lambda}}$ is a homological trivial family. By Lemma 2.10, there is a diffeomorphism

$$h:(\widetilde{X}_q,\widetilde{\omega}_q)\longrightarrow(\widetilde{X}_{q_*},\widetilde{\omega}_{q_*})$$

preserving the cohomology markings. Define

$$\alpha_\delta:=[q\mapsto h^*\Psi_q^*\widetilde{\omega}_q]\in\pi_{2r-1}(\mathcal{S}_{\widetilde{\omega}},\widetilde{\omega}).$$

4.3. **Evaluation of q-invariants and proof of Theorem A.** In this section, we evaluate $q_{\delta-\sum_{i\in I_k}e_i}(\alpha_{\delta'})\in\mathbb{Z}_2$ for $\delta,\delta'\in\Delta_{X,k}$, and prove the main theorem.

Using $\Psi_q\circ h:\widetilde{X}\to\widetilde{X}_q$, we pull the complex structures and Kähler forms back to $\widetilde{X}$, keeping the notation $\tilde{J}_q$ and $\widetilde{\omega}_q$ for $q\in B_\delta$.

Lemma 4.4. *Set $\epsilon=\delta-\sum_{i\in I_k}e_i$, then*

$$Q_\epsilon(\alpha_\delta)=\mathrm{FSW}_{\mathfrak{s}_\epsilon}\big(\widetilde{X}\times B_\delta,\mathfrak{T}_\rho(\widetilde{\omega}_q,\widetilde{J}_q)\big)=1\in\mathbb{Z}_2$$

for all sufficiently large ρ.

Proof. For simplicity, we write $\alpha=\alpha_{\kappa,\boldsymbol{\lambda}}$ and $g_q=\widetilde{\omega}_q(\cdot,\widetilde{J}_q\cdot)$. Both κ and each exceptional class e_i are contained in $H^{1,1}(\tilde{X}_q;\mathbb{R})$, so α and $[\widetilde{\omega}_q]$ both belong to $H^{1,1}(\widetilde{X}_q;\mathbb{R})$ for every $q\in B_\delta$. Since $D_\delta\subset\mathcal{M}_\kappa$, κ is contained in the positive cone of $H^{1,1}(\tilde{X}_q;\mathbb{R})$, and

$$\alpha^2>0,\qquad[\widetilde{\omega}_q]^2>0,\qquad\alpha\cdot\kappa=\kappa^2>0,\qquad[\widetilde{\omega}_q]\cdot\kappa>0.$$

Then by the Hodge index theorem, α and $[\widetilde{\omega}_q]$ lie in the same component of the positive cone of $H^{1,1}(\widetilde{X}_q;\mathbb{R})$. Consequently,

$$\alpha\cdot[\widetilde{\omega}_q]>0\quad\text{ for each }q\in B_\delta.$$

By [Lemma 4.15, [7]], $Pr(\mathfrak{T}_\rho(\widetilde{\omega}_q,\widetilde{J}_q))$ will be contained in $\mathbf{K}_{-\alpha}(\widetilde{X})$ for all sufficiently large ρ. By Lemma 3.4, $\{\mathfrak{T}_\rho(\widetilde{\omega}_q,\widetilde{J}_q)|q\in B_\delta\}$ is an admissible extension of its restriction to S_δ, and the first equality holds.

Clearly, there is a curve $\tilde{C}_0$ in the class $\delta-\sum_{i\in I_k}e_i$ contained in the surface $\tilde{X}_{q_0}$. Conversely, we suppose that $D\subset\widetilde{X}_q$ is an effective divisor in class ϵ, where $q=(u,x_1,\dots,x_n)$. Let $\pi:\widetilde{X}_q\to X_u$ be the natural projection, then $\pi_*(D)$ is an effective divisor of X_u in class δ. Since $D_\delta\cap H_\delta=\{u_0\}$, we must have $u=u_0$. When X is a K3 surface, its pushforward is C_0; if X is a torus, the pushforward is a smooth fiber of the elliptic fibration (4.2). Denote it by C, then $x_i\in C$ for each $i\in I_k$. In the torus case, $x_1=Z(u_0)\in C$ implies that $C=C_0$. By the construction of B_δ, we can see that $q\in B_\delta\bigcap\mathcal{H}_\delta$, hence $q=q_0$ and $D=\widetilde{C}_0$. By [30][8], $\bigcup_{q\in B_\delta}\mathcal{M}^{\mathfrak{s}_\epsilon}_{\mathfrak{T}_\rho(\widetilde{\omega}_q,\tilde{J}_q)}$ consists of a unique point, which is corresponding to $\tilde{C}_0$ in $\tilde{X}_{q_0}$.

It remains to prove regularity of the point in $\bigcup_{q\in B_\delta}\mathcal{M}^{\mathfrak{s}_\epsilon}_{\mathfrak{T}_\rho(\widetilde{\omega}_q,\tilde{J}_q)}$. Set $a=0$ for K3 surfaces and $a=1$ for tori. Over $B^{(a)}=D_\delta$, we take the following family:

$$\mathcal{X}^{(a)}=\begin{cases}\mathcal{X}_\kappa|_{D_\delta}, & X\text{ is a K3 surface},\\ \mathrm{Bl}_{Z(D_\delta)}\mathcal{X}_\kappa|_{D_\delta}, & X=T^4.\end{cases},\quad C_a=\begin{cases}C_0, & X\text{ is a K3 surface},\\ \text{proper transform of }C_0, & X=T^4.\end{cases}$$

By [§ 8, [27]] and [§ 9, [28]], $res \circ \mathrm{KS} : T_0 D_\delta \to H^1(C_a, N_{C_a/\mathcal{X}_{u_0}^{(a)}})$ is surjective, respectively.

For $a < j \leqslant k$, we take jointly holomorphic maps

$$F_j : D_\delta \times \Delta_j \longrightarrow \mathcal{X}_\kappa|_{D_\delta}, \qquad p_\kappa \circ F_j = \mathrm{pr}_{D_\delta}, \qquad F_j(u_0, 0) = x_j(q_0),$$

such that $\{F_j(u_0, z) | z \in \Delta_j\}$ is transverse to C_0 for each j and the images of F_j are disjoint to each other, and disjoint from the zero section Z in the torus case. Here, $x_j(q_0)$ is the coordinate of q_0. By shrinking Δ_j if necessary, we assume that $F_j(u_0, \cdot)^{-1}(C_0) = \{0\}$. For each $i > k$, choose a holomorphic section $s_i : D_\delta \to \mathcal{X}_\kappa|_{D_\delta}$ with $s_i(u_0) = x_i(q_0)$. By shrinking the image of s_i, we may again assume that the images are pairwise disjoint, disjoint from F_j and, for tori, the zero section Z. Set

$$P_\delta = D_\delta \times \prod_{i=a+1}^{k} \Delta_i, \qquad B^{(j)} = D_\delta \times \prod_{i=a+1}^{j} \Delta_i.$$

The holomorphic maps F_i, the sections s_i, together with the section Z in the torus case, determine a holomorphic map

$$\Psi : P_\delta \longrightarrow \mathcal{P}_{D_\delta}, \qquad (u, \mathbf{z}) \longmapsto (u, x_1(u, \mathbf{z}), \ldots, x_n(u, \mathbf{z})),$$

$$x_i(u, \mathbf{z}) = \begin{cases} F_i(u, z_i), & a < i \leqslant k, \\ Z(u), & X = T^4,\ i = 1, \\ s_i(u), & i > k. \end{cases}$$

In particular, $\Psi(u_0, 0) = q_0$. For $a + 1 \leqslant j \leqslant k$, suppose inductively that the holomorphic family

$$p^{(j-1)} : \mathcal{X}^{(j-1)} \longrightarrow B^{(j-1)}$$

has been constructed, starting from $\mathcal{X}^{(a)}$ as above. Its fiber over $(u, z_{a+1}, \ldots, z_{j-1})$ is obtained from blowing up each fiber of D_δ by successively blowing up the points

$$F_{a+1}(u, z_{a+1}), \ldots, F_{j-1}(u, z_{j-1})$$

simultaneously (also blowup along $Z(u)$ if X is torus). Let

$$\Pi_{j-1} : \mathcal{X}^{(j-1)} \longrightarrow B^{(j-1)} \times_{D_\delta} (\mathcal{X}_\kappa|_{D_\delta})$$

be the corresponding blowdown map. Set $\mathcal{Y}^{(j)} = B^{(j)} \times_{B^{(j-1)}} \mathcal{X}^{(j-1)}$, since the image of F_j is disjoint from the images of $F_{a+1}, ..., F_{j-1}$ (and $Z(u)$ if $X = T^4$), the lift of F_j under Π_{j-1} defines a holomorphic section

$$\sigma_j : B^{(j)} \longrightarrow \mathcal{Y}^{(j)},$$

and $p^{(j)}$ is defined as follows

$$p^{(j)} : \mathcal{X}^{(j)} = \mathrm{Bl}_{\sigma_j(B^{(j)})} \mathcal{Y}^{(j)} \longrightarrow B^{(j)}.$$

Let $X_0^{(j)}$ be the central fiber of $\mathcal{X}^{(j)}$, and let $C_j \subset X_0^{(j)}$ be the proper transform of C_a. Since the image of F_j are disjoint from each other (and $Z(u)$ when $X = T^4$), and $\{F_j(u_0, z) | z \in \Delta_j\}$ is transverse to C_0 for each j, then the lift of $\{F_j(u_0, z)\}$ remains

transverse to C_{j-1}. Together with the surjectivity of $res \circ \mathrm{KS} : T_0 D_\delta \to H^1(C_a, N_{C_a/\mathcal{X}_{u_0}^{(a)}})$, we can apply Theorem 3.14 at each step and get the surjectivity of

$$\mathrm{res} \circ \mathrm{KS}_j : T_0 B^{(j)} \longrightarrow H^1(C_j, N_{C_j/X_0^{(j)}})$$

for every $a \leqslant j \leqslant k$.

For each $i > k$, s_i are disjoint from C_0, blowing up $\mathcal{X}^{(k)}$ along the lifts of s_i, the blowdown map of each fiber is a biholomorphism near C_k and hence preserve its normal bundle and the restriction of the Kodaira–Spencer map. Since the image of s_i are pairwise disjoint and disjoint from F_j and, for tori, $Z(u)$, the resulting family can be considered as a pullback of the blowup family (2.16) by Ψ. Therefore, the surjectivity obtained above implies that

$$\mathrm{res} \circ \mathrm{KS} : T_{q_0} \mathcal{P}_{D_\delta} \longrightarrow H^1(\widetilde{C}_0, N_{\widetilde{C}_0/\widetilde{X}_{q_0}})$$

is surjective.

Finally, the restriction of $\widetilde{p} : \widetilde{\mathcal{X}} \to \overline{\mathcal{P}}_\kappa$ to a neighborhood of q_0 in $\mathcal{H}_\delta$ is a deformation of the pair $(\widetilde{X}_{q_0}, \widetilde{C}_0)$, hence $T_{q_0}\mathcal{H}_\delta \subseteq \mathrm{Ker}(\mathrm{res} \circ \mathrm{KS})$. Since

$$h^1(\widetilde{C}_0, N_{\widetilde{C}_0/\widetilde{X}_{q_0}}) = \mathrm{codim}_{\mathbb{C}}(\mathcal{H}_\delta, \mathcal{P}_{D_\delta}) = r,$$

and B_δ is transverse to $\mathcal{H}_\delta$, we conclude that

$$\mathrm{res} \circ \mathrm{KS}|_{B_\delta} : T_{q_0} B_\delta \xrightarrow{\cong} H^1(\widetilde{C}_0, N_{\widetilde{C}_0/\widetilde{X}_{q_0}}).$$

By [8][13], the solution in $\bigcup_{q \in B_\delta} \mathcal{M}^{\mathfrak{s}_\epsilon}_{\mathfrak{T}_\rho(\tilde{\omega}_q, \tilde{J}_q)}$ represented by $\widetilde{C}_0$ is regular, and

$$Q_\epsilon(\alpha_\delta) = \mathrm{FSW}_{\mathfrak{s}_\epsilon}\big(\widetilde{X} \times B_\delta, \mathfrak{T}_\rho(\widetilde{\omega}_q, \widetilde{J}_q)\big) = 1 \in \mathbb{Z}_2.$$

□

Lemma 4.5. $Q_{K-(\delta-\sum_{i\in I_k} e_i)}(\alpha_\delta) = Q_{\sum_{j\in\{1,\dots,n\}\setminus I_k} e_j + 2\sum_{i\in I_k} e_i - \delta}(\alpha_\delta) = 0.$

Proof. The admissibility argument in Lemma 4.4 applies equally to $\mathfrak{s}_{K-\epsilon}$ for sufficiently large ρ, and solutions in $\bigcup_{q\in B_\delta} \mathcal{M}^{\mathfrak{s}_{K-\epsilon}}_{\mathfrak{T}_\rho(\tilde{\omega}_q, \tilde{J}_q)}$ correspond to effective divisors in class $K - \epsilon$. By [Theorem 1.3, [29]], if $Q_{K-(\delta-\sum_{i\in I_k} e_i)}(\alpha_\delta) \neq 0$, then there is $q \in B_\delta$ and a curve $\tilde{C} \subset \tilde{X}_q$ such that $\mathrm{PD}[\tilde{C}] = K - (\delta - \sum_{i\in I_k} e_i)$. As a result, $-\delta$ is an effective class in the minimal model of $\tilde{X}_q$. However, the minimal model of $\tilde{X}_q$ is X_u, where $u \in \mathcal{M}_\kappa$. This contradicts the fact that $\langle \kappa, \delta \rangle > 0$.

□

By the local finiteness of hyperplanes in $\mathcal{I}_X(S)$ and the construction of $D_{\delta'}$, $\delta \notin H^{1,1}(\tilde{X}_q, \mathbb{R})$ for each $\delta' \in \Delta_{X,k}\backslash\{\delta\}$ and $q \in B_{\delta'}$. For either class $\epsilon = \delta - \sum_{i\in I_k} e_i$ or $K - \epsilon$, the same admissibility argument applies to the Kähler family over $B_{\delta'}$. Since δ' is not of $(1,1)$-type in each fiber of B_δ, and corresponding moduli spaces are empty by

[Theorem 1.3, [29]]. Thus, $Q_{\delta-\sum_{i\in I_k}e_i}(\alpha_{\delta'}) = Q_{K-(\delta-\sum_{i\in I_k}e_i)}(\alpha_{\delta'}) = 0$ if $\delta' \neq \delta$. Thus we have

$$q_{\delta-\sum_{i\in I_k}e_i}(\alpha_{\delta'}) = Q_{\delta-\sum_{i\in I_k}e_i}(\alpha_{\delta'}) = \begin{cases} 1, & \delta' = \delta, \\ 0, & \delta' \neq \delta. \end{cases} \tag{4.5}$$

Set

$$G = \mathrm{Ker}\Big(\pi_{2r-2}(\mathrm{Symp}_{st}(\widetilde{X},\widetilde{\omega})) \longrightarrow \pi_{2r-2}(\mathrm{Diff}_0(\widetilde{X}))\Big).$$

For $\delta \in \Delta_{X,k}$, write $\epsilon_\delta = \delta - \sum_{i\in I_k} e_i$ and $\beta_\delta = \partial_*\alpha_\delta \in G$. By Lemma 3.12 and (4.5),

$$\bar{q}_{\epsilon_\delta}(\beta_{\delta'}) = \begin{cases} 1, & \delta = \delta', \\ 0, & \delta \neq \delta'. \end{cases}$$

Consequently, for every finite subset $F \subset \Delta_{X,k}$, the homomorphism

$$(\bar{q}_{\epsilon_\delta})_{\delta\in F} : G \longrightarrow \bigoplus_{\delta\in F} \mathbb{Z}_2$$

is surjective. Since $\Delta_{X,k}$ is infinite, G is not finitely generated, proving Theorem A.

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